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A single dynamical property can account for the capacity to learn, from artificial networks to the mammalian brain.

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SUMMARY STATEMENT

Every brain must adapt to an unpredictable world, yet individuals differ in how readily they learn. Theoretical work suggests that learning is fastest when a system, whether biological or synthetic, is initialized in a state close to instability - i.e., near *criticality* - because critical dynamics are imbued with a diverse repertoire of patterns and multi-scale correlations. Here, we empirically estimate distance to criticality in the brain and show that it predicts the rate of adaptability underlying learning, neuronal tuning, and general intelligence. In mouse motor cortex, proximity to criticality forecasts learning rate of two future complex tasks: prey capture hunt and ladder crossing. In contrast, distance to criticality predicted neither an animal's naive ability nor its asymptotic skill — isolating the rate of learning itself. In visual cortex of young ferrets, proximity to criticality predicts how strongly experience reshapes neural tuning. In human frontal cortex, it correlates with general cognitive ability. A minimal recurrent network model reproduced these results and offers a mechanism: proximity to criticality defines the timescale over which a system can learn from its past experiences, directly setting the rate of learning. A single dynamical property can account for the capacity to learn, from artificial networks to the mammalian brain.

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Introduction

What is the basis of an individual's durable intelligence? What is the biological foundation of the capacity to learn? Any complex system capable of adapting to an unpredictable environment should benefit from exploring as large a space of possible solutions as quickly as possible. This principle is well established in artificial intelligence and machine learning, where solutions are found quickly only when diverse patterns of activation spread across nodes and layers without saturating the system[1–5]. Does learning in biological systems obey the same principle? If so, a brain that generates a larger repertoire of activity patterns should be the more adaptable and thus learn faster.

How should brain dynamics be configured to achieve such an adaptable state? Converging insights from machine learning, neuroscience, and physics point to a compelling answer. In machine learning, learning proceeds fastest when a network is initialized near the boundary between two dynamical regimes — stability and chaos — a state variously termed the "edge of chaos"[6–9] or "criticality"[1,4,10–13]. This holds whether learning is driven by conventional gradient-descent backpropagation[1,3–5,14] or by more biologically realistic rules[2]. Moreover, keeping a network near criticality throughout learning, rather than only at initialization, further improves the quality of what it learns[3,14]. This marginally stable, critical state is what endows the system with the large repertoire of activity patterns thought to accelerate learning[15–18]. The reason is rooted in the fundamental physics of criticality: by renormalization group theory[19,20], scale invariance in interacting systems arises *exclusively* at critical fixed points — other arrangements of a system generate correlations that disappear across scales[21]. It is therefore only at criticality that a system can generate patterns of every scale, from small to large and fleeting to persistent. This informs our central hypothesis: living organisms learn faster the closer they are to criticality.

Congruent with a role in the brain's general ability to produce adaptive behavior, near critical dynamics are evident across species and are consistent with an endpoint of homeostatic plasticity[10,22–26]. As biology cannot tune to a point, homeostatically-controlled physiologic parameters, such as body temperature and blood pH[27], are actively maintained within a viable range. This must also be true of similar set-points in brain dynamics. For this reason, a brain cannot "be critical"; instead, neuronal activity should be described according to its *distance* from a critical boundary[28]. As a population is tuned further and further from criticality, the scale-invariant range shrinks[29,30], and computational capacity diminishes[16,31–34]. Additionally, when considering homeostatically controlled physiological features, population-level variance is often larger than individual-level variance. Plainly, there is truth to the idea that some people run hotter than others[35]. It stands to reason that, as a homeostatic set-point, proximity to criticality should be no different. Specifically, should healthy brain circuits be actively maintained near criticality[22,23], some circuits and/or individuals may be expected to be consistently closer (or further) than others. If true, and if proximity to criticality establishes the capacity for adaptive computation, then proximity to criticality should both determine the facility with which a brain can learn, and account for persistent inter-individual differences in the rate of learning and adaptation.

Results

Isocortical proximity to criticality predicts future learning

To directly test the hypothesis that baseline feature(s) of an animal's neural dynamics account for inter-individual differences in learning (Fig. 1A), we continuously tracked the activity of ensembles of putative excitatory single neurons (regular spiking units; RSUs[36–38]) in right primary motor cortex (M1) of adult wild-type C57BL/6 mice ($N = 18$) for a total of 13 to 14 days (312-336 h) per

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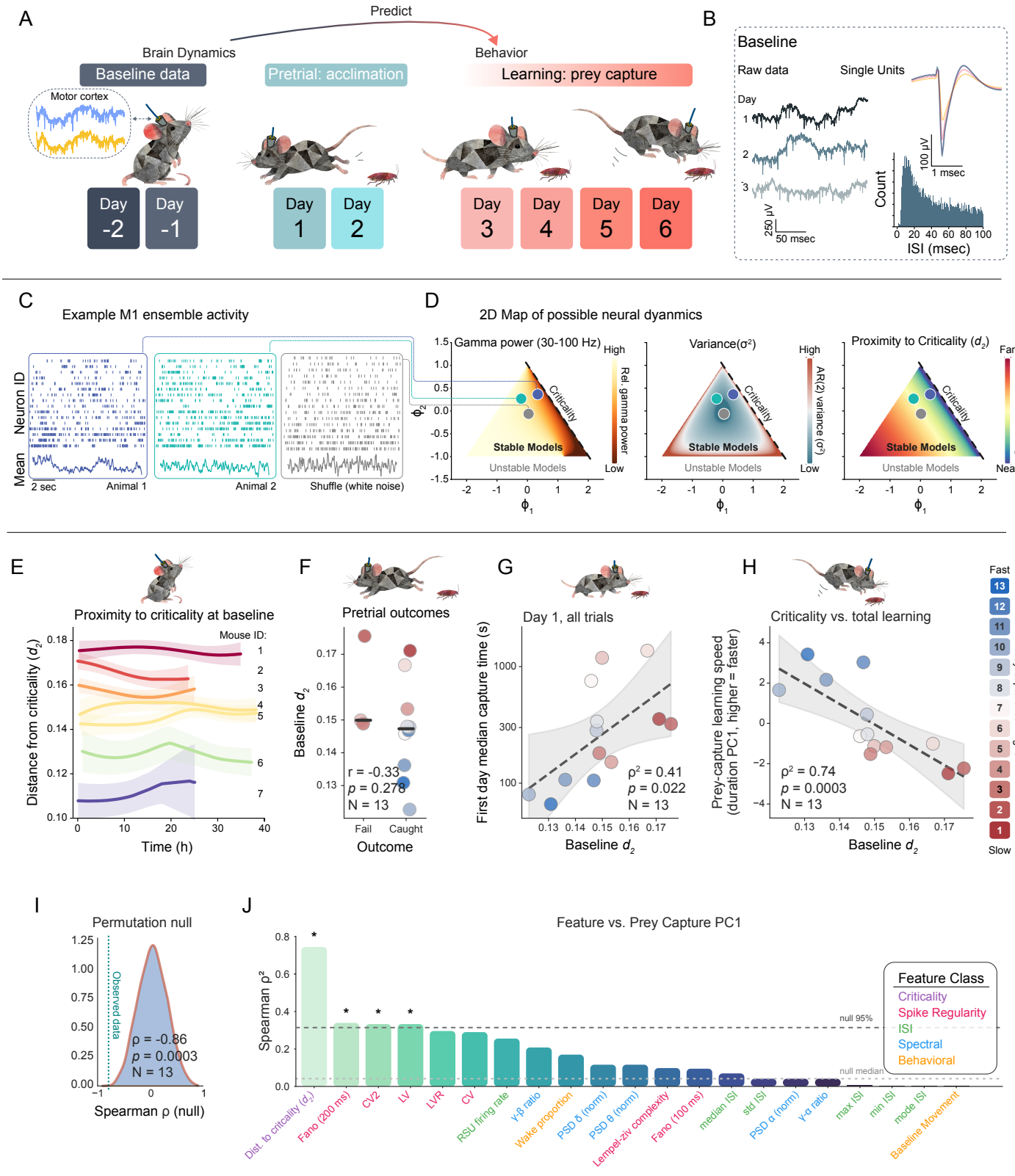
animal (6,326 RSUs total, mean = 34 ± 13.9, minimum = 15, maximum = 71) (Fig. 1B). This 13-14 d continuous recording comprised a 24-48 h baseline of spontaneous behavior in the home cage, followed by 6 d in each of two motor learning tasks, prey capture and ladder-crossing (Fig. S1). Animals were randomly assigned to ladder first or prey capture first. All animals included in these experiments were wild type (WT) and thus expected to exhibit modest variability in cognitive performance relative to diseased animals or a genetically heterogeneous population. We attempted to forecast future learning in each task from a suite of neural and behavioral measures collected before animals had any experience with either paradigm (Fig. 1C,D). During the baseline period, three arousal states were labeled based on standard polysomnography (local field potential spectral power, video)[23,39]: wake, NREM sleep, and REM sleep. Here, unless otherwise mentioned, all neural features are examined exclusively during waking epochs.

Any sample of neural data can be described in terms of a wide range of metrics, including firing rate, variance, spectral power, and more. Each of these measures is likely related to the tendency to generate the diverse repertoire of activity patterns needed for fast learning. However, a direct test of our hypothesis requires a direct measurement of proximity to criticality. For this, we use d_2, a recently developed method for empirically estimating proximity to criticality based on univariate time series[40]. Leveraging a renormalization group theory of autoregressive models, d_2 quantifies the extent to which a given sample reflects scale-invariant fluctuations (Fig. 1D, S2). Concretely, purely sinusoidal activity is defined by a single time scale, while critical dynamics are defined by correlations at all scales. In practice, d_2 is calculated by first fitting a model to observed data and then measuring the distance from the best fit model to the nearest model whose activity is scale invariant (i.e., distance to the dashed border in Fig. 1D). Briefly, we converted the spike times of single unit ensembles into a population spike count timeseries. We fit a 10th order autoregressive (AR) model to the time series, yielding 10 AR coefficients, thus specifying a point in a 10-dimensional parameter space of all possible 10th order AR models (example for AR(2) model in Fig. 1D). In this space, d_2 is the minimum Euclidean distance between the best-fit model point and the hyperplane formed by the set of critical models, i.e., 10th order models that generate autocorrelations across all scales. In our animals, the mean motor cortex d_2 was 0.1494 ± 0.000489 ($N = 18$). For comparison, we shuffled spike times to maintain population activity rate while destroying temporal correlations, which provides a quantitative example of d_2 for dynamics 'very far' from criticality (Fig. 1C). This produced mean $d_{2\,shuffle}$ of 0.3166 ± 0.00006, close to the analytical solution for white noise with order 10 (d_2 for white noise $= \frac{1}{\sqrt{\text{AR order}}}$) of $d_2 = 0.32$. Practically, white noise defines the upper limit of d_2 short of invoking an unstable regime (for example, seizure). Note that a wide range of physiologically-plausible dynamics are well described by AR models — the same AR landscape can thus be evaluated in terms of any descriptor of neural activity; Fig. 1D shows the same AR(2) landscape as a function of gamma power, variance, and d_2. This offers a direct comparison of alignment across metrics (Fig. 1D). Any movement in this landscape invariably entails changes in multiple features.

Isocortical d_2 reliably differentiates animals

If distance from criticality is homeostatically regulated, it is, by definition, actively maintained within some effective range[22]. While animals could, in theory, drift without constraint within this range, many homeostatically controlled parameters are comparatively tight at the level of the individual organism[37,41–43]. As a result, an individual may be typified as "running hot" relative to another. Whether distance to criticality can reliably identify an animal is unknown, yet this is a logical predicate for our hypothesis that it comprises a durable feature of cognition. To assess

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Figure 1: Proximity to criticality predicts prey capture learning rate. **(A,B)** Experimental design: ensembles of single units in primary motor cortex were recorded for two baseline days. Baseline, single unit neural activity was used to predict various aspects of a subsequent six day prey capture learning protocol. Pretrial acclimation allowed mice to acclimate to cockroaches. Next, mice completed 24 food-deprived hunting trials spread across four days. **(C)** Example ensemble rasters and population activity range from richly structured near criticality (low d_2) to progressively flatter and more noise-like as d_2 grows, with a spike-time-shuffled control approximating white noise. **(D)** Autoregressive (AR) models reproduce physiologically-relevant neural dynamics. A second-order AR model is defined by two variables, ϕ_1 and ϕ_2, allowing a 2D visualization of all possible models. The central triangle comprises the set of models with stable dynamics. These can be quantified in terms of, e.g., gamma band power **(left)**, variance **(center)**, and the shortest distance to a critical model **(right)**. Note that, in this 2D space, the upper right boundary of stable models (dashed line) is the set of models that generate scale-invariant, critical activity. **(E)** At baseline, proximity to criticality (measured as d_2 during waking epochs) is stable within individual animals over time. **(F)** Baseline d_2 does not predict whether or not mice consume the pretrial cockroach ($r = -0.33$, $p = 0.278$). **(G)** In contrast, on the first day of trials (i.e., the first opportunity to refine behavior as a function of experience), baseline d_2 predicts median capture time - where mice closer to criticality at baseline are faster hunters ($\rho^2 = 0.41$, $p = 0.022$, dashed line, fit; shaded band, 95% CI). **(H)** Across all 24 trials over four days, d_2 at baseline predicts a four-measure composite of prey-capture learning ($\rho^2 = 0.74$, $p = 0.00033$, dashed line, fit; shaded band, 95% CI). F-G report the $N = 13$ mice that hunted. Individuals are colored by their learning ranking (H, right). **(I)** A permutation null confirms that the learning effect lies outside chance. Observed value, dashed line; $p = 0.0003$). **(J)** The predictive value of twenty one features measured during the baseline. Features include neural, spectral, and behavioral metrics. In addition to d_2, three spike-regularity measures (Fano-200 ms, CV2, LV) clear the 95th-percentile permutation null (dashed line; asterisks).

this, we examined the source of variance in d_2 during the baseline (Fig. 1E). Considered across $N = 18$ animals, baseline d_2 spanned a 1.6-fold range of setpoints (per-animal means 0.111–0.176; between-animal $SD = 0.015$) (Fig. 1F). Individual 5-min estimates of d_2 were themselves noisy, such that animal identity accounted for only $\sim 30\%$ of the variance in single windows (one-way random-effects $ICC(1,1) = 0.30$, 95% CI 0.15–0.42), the remainder reflecting fast within-animal fluctuation (Fig. S3). This within-animal variance, however, largely averaged out ($ICC(1,k) = 0.94$). Each animal's setpoint (baseline mean) was a stable and reproducible feature. Between-animal differences in setpoint exceeded chance even after accounting for the autocorrelation of successive windows (one-way ANOVA on independent 3 h blocks, $F(17,79) = 12.6$, $p = 1 \times 10^{-4}$ by label permutation). Furthermore, an animal's setpoint estimated from the first half of its baseline predicted the setpoint measured over the second, non-overlapping half (test–retest $r = 0.83$, 95% CI 0.58–0.94; Spearman–Brown full-length reliability = 0.91). Consistent with a durable individual trait, baseline d_2 alone was sufficient to distinguish animals (Fig. S3). Pairwise discriminability rose monotonically with the difference in setpoint (Spearman $\rho = 0.94$ between AUC and setpoint separation, $p < 10^{-69}$), reaching a median AUC of 0.68 from a single 5-min window and 0.89 with ~ 2.7 h of data, and a cross-validated classifier identified the animal from d_2 at twice the chance rate across all 18 individuals ($p = 5 \times 10^{-4}$). Thus, although distance to criticality fluctuates from moment to moment, its setpoint is a reliable individual trait. Some animals sit consistently closer to criticality than others.

In the five animals implanted in both M1 and V1, simultaneously recorded d_2 shared only a modest fraction of its moment-to-moment variance across the two regions (median within-animal Spearman $\rho = 0.33$) and held distinct regional set-points, indicating that proximity to criticality is regulated locally within a circuit rather than imposed by a single brain-wide metabolic or global

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physiological variable (Fig. S4).

Isocortical proximity to criticality forecasts prey capture learning rates

Mathematically, a system's proximity to criticality constrains the transmission of information through time and across space[31]. Given a fixed number of neurons, distance from criticality directly limits the extent and rate of learning. On this basis, we reasoned that animals tuned closer to criticality should learn more quickly than those tuned further. To test this, we asked whether animal-level set-points in proximity to criticality during the baseline period could forecast learning rate in either of two six-day tasks: prey-capture and ladder-crossing. We first examined prey-capture, in which mice that were bred and raised in an academic vivarium were introduced to red runner cockroaches (*Shelfordella lateralis*). All mice were acclimated to red runners during an overnight pre-trial exposure (days 1-2) with *ad lib* food access, followed by food-deprived hunting trials (days 3-6). Five of 18 mice never hunted, and were thus excluded from these analyses, and whether a mouse hunted or not was independent of d_2 (S5A).

Importantly, learning is the means by which an animal transitions from its starting point to its end point via experience. In the context of our prey capture task, whether an animal is initially frightened or eventually athletic should be largely independent of the intervening period of plastic refinement. Consistent with this, in wild type animals, distance to criticality at baseline carried no information about how animals responded to the pretrial red runner. Note that, during the pre-trial, mice acclimated to a novel animate object, overcoming neophobia and, over hours, most slowly realized the cockroach could be treated as a food source. Of the hunters (13/18), 77% (10/13) caught and consumed the pretrial red runner, which was unrelated to baseline d_2 (point-biserial $r^2 = 0.11$, $p = 0.28$; Fig. 1F). Pre-trial capture time ranged from 0.12 to 11.8 h and was also not predicted by baseline d_2 ($\rho^2 = 0.006$, $p = 0.83$, Fig. S5B). Thirty six hours after the pre-trial night, food-deprived hunting trials began. Mice completed a total of 24 hunting trials (six red runners / day for four days). The first hunting trial is unique in that it arises with a motivation to pursue food but no prior hunting experience, and thus reflects an animal's starting point for learning. In line with our hypotheses, murine proximity to criticality did not predict performance on the first trial ($\rho^2 = 0.143$, $p = 0.23$; Fig. S5C), suggesting that the dynamical regime of M1 indicates nothing about innate hunting prowess or motivation. Following the first trial, all animals exhibited robust improvements in hunting performance even on Day 1: every hunter (13/13) was faster on a subsequent Day-1 trial than on its naive first trial (Fig. S5D). Median capture time dropped from 761 (Trial 1) to 95 s (fastest capture time on Day 1) across mice, an 88% reduction (Wilcoxon signed-rank $p = 0.000244$).

When considering the first opportunity for learning (Day 1 trials), mice closer to criticality at baseline were more effective in learning to hunt quickly than those further from criticality. In contrast to the very first trial, baseline d_2 predicted Day 1 median capture time ($\rho^2 = 0.41$, $p = 0.02$, Fig. 1G). All animals continued to shorten capture time across the 24 trials (13/13 with a negative trial-vs-time correlation, median Spearman $\rho = -0.70$, sign-test $p = 1.2 \times 10^{-4}$), with gains decelerating toward an asymptote within the first ∼10–15 trials (exponential $\tau \approx 6$ trials; day-over-day reduction falling from 223 s to 14 s, paired Wilcoxon $p = 5 \times 10^{-4}$) (Fig. S5D). To avoid over-reliance on any one metric, we quantified learning along three axes related to two independent methods (breakpoint analysis and exponential fitting of capture times, Fig. S5E-G). Individually, the three metrics were highly concordant (Kendall's $W = 0.88$, Fig. S5F), and we constructed a composite learning-rate metric as the first principal component across the three measures, which captured 89% of their shared variance. Considered over the four day, 24 trial protocol, baseline proximity to criticality was strongly correlated with this composite (Spearman

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$\rho^2 = 0.74$, $p_{\text{perm}} = 0.0004$, $N = 13$; Fig. 1H): mice closer to criticality at baseline — i.e. with lower d_2 — acquired the skill more rapidly, with baseline d_2 alone explaining $\sim 74\%$ of the variance in learning rate. This relationship did not depend on any single animal (leave-one-out Spearman $\rho \in [-0.888, -0.832]$), and each constituent metric tracked baseline d_2 on its own (Fig. S5G). We quantify the d_2–learning relationship with Spearman's ρ because hunting performance is a bounded, saturating outcome. Once learning is complete (i.e., animals have reached their behavioral asymptote), d_2 was meaningless as a predictor: the median time-to-subdue over each animal's final ten trials (22–220 s across mice) did not covary with baseline d_2 (Spearman $\rho = +0.36$, $p = 0.23$), nor did the fitted asymptote of the capture-time learning curve ($\rho = -0.29$, $p = 0.34$), nor the mean late-phase capture latency ($\rho = +0.24$, $p = 0.44$) (Fig. S6A-B). Baseline d_2 likewise carried no signature of athletic capacity: mean, peak, and total speed during hunts did not significantly covary with distance to criticality (all $p \geq 0.08$; data not shown). These findings, paired with our findings during the learning phase, reinforce the conclusion that proximity to criticality specifically predicts rate of experience-dependent improvement in hunting performance, without any relationship to the point where learning started or its ultimate result: in the vocabulary of machine learning, a learning rate.

While the strength of this relationship, particularly in a freely behaving neurobiological context, is perhaps surprising, we reasoned that it may not be unique[44,45]. To assess this, we analyzed the predictive power of 20 additional neural, spectral, and behavioral features from the baseline period (Fig. 1J). Only 3/20 features rose above the 95th-percentile of the Spearman ρ^2 permutation null, each a measure of spike regularity (200 ms fano factor[46], local LV[47], and CV2[48]). d_2 ($\rho^2 = 0.74$) outperformed the best of these (Fano-200 ms, $\rho^2 = 0.34$) by more than 40 points of explained variance. To understand how d_2 relates to other features of neural activity, we computed each metric in closed form across the space of stable second-order (AR(2)) models and compared it to d_2 in the same ϕ_1–ϕ_2 plane (Fig. 1D, Fig. S2). Within any local region of this space, nearly every standard descriptor of spiking covaries with d_2. However, these relationships are often not monotonic, and don't hold across the entire map (Fig. S2). Taken together, these data support the hypothesis that initial proximity to criticality determines of the rate of subsequent learning in the brain. However, prey capture could appear purpose-built for confirming the critical brain hypothesis[10,49,50]. The task is complex, integrative, and ever changing, matching the defining features of critical dynamics themselves. This raises the question of whether criticality might not be predictive of the many more repetitive and low dimensional motor patterns that comprise much of daily behavior[13]. Accordingly, we examined learning in a second, six-day task in the same mice where mastery is defined by precision and stereotypy: crossing a 10-rung ladder.

M1 proximity to criticality predicts learning of a stereotyped motor task: ladder crossing

Briefly, 18/18 mice spent six days living in a custom homecage whose two halves were separated, raised, and connected by a suspended ladder equipped with capacitive touch sensors (Fig. 2A). Ad lib food was available on one side, and water on the other. Beyond the imposed necessity to cross, animals engaged extensively with the ladder, crossing a mean of 649 times per 24 h (Minimum 60 crosses / day, maximum 1560 crosses / day, median 622 crosses / day). Two mice were excluded due to equipment malfunction. The capacitive touch time series made quantification and classification of crossing behaviors robust. Consistent with a lack of either prior ladder experience or a hard-wired ladder-walking circuit, early crossing attempts were hesitant, irregular, and slow. Eventually, all mice exhibited a set of readily distinguishable cross types, including stall-and-sniff, u-turns, walks, and highly stereotyped trots (Fig. 2B).

In the six days of ladder crossing, two categories of learning were readily discernible: an early

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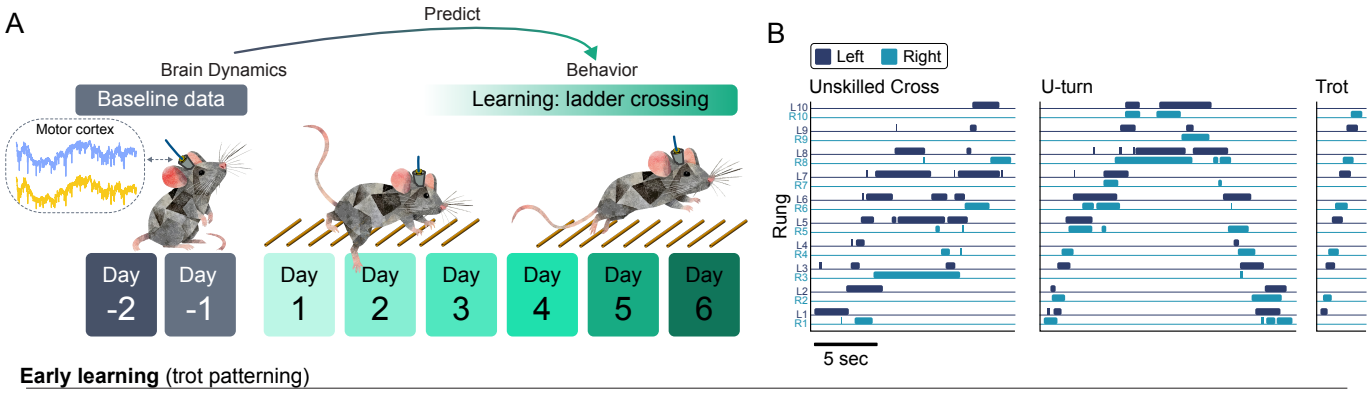
phase in which mice learned to trot (pattern emergence and consolidation; Fig. 2C), and a late phase in which mice exhibited their fastest successful crosses (refinement and further improvement). Early learning was evident in the first 100 ladder interactions, and was well described by multiple measures, such as, e.g., the number of trots in this set and the number of crosses to achieve the nth trot. While all were highly concordant (Kendall's $W = 0.914$), we calculated a composite as the first principal component across four measures (Fig. S7A-C). Within the first ∼100 ladder interactions, all 16 animals learned to successfully trot, but exhibited a spread in early learning rates (for example, a 29-fold range in n crossings to the 2nd trot).

While ladder trotting is repetitive, low in variance, and largely fixed in its spatial and temporal aspects, we reasoned that the ability of a naive neurobiological network to explore a space of possible solutions for the behavior would nonetheless be limited by the initial proximity to criticality. To test this, we asked if baseline proximity to criticality was sufficient to forecast the rate of early ladder learning (recall that 8 animals went from baseline immediately to the ladder, while 10 first completed a 6 d prey capture paradigm). Consistent with prey capture learning, baseline proximity to criticality strongly predicted the rate at which each mouse learned to trot ($\rho^2 = 0.612$, $p_{\text{perm}} = 0.0003$, $N = 16$; Fig. 2D). The effect was robust to leaving out any single animal (leave-one-out Spearman $\rho \in [-0.882, -0.732]$) and held for each early-learning metric individually (Fig. S7B,C). We again examined the predictive power of a panel of 20 additional baseline features, none of which reached the 95th-percentile of the permutation null ($\rho^2 = 0.250$) (Fig. 2E). The closest feature, LVR (local variation of rate; $\rho^2 = 0.182$), was not significant but is noteworthy as the closest measures to d_2 in prey capture also quantified variance[51].

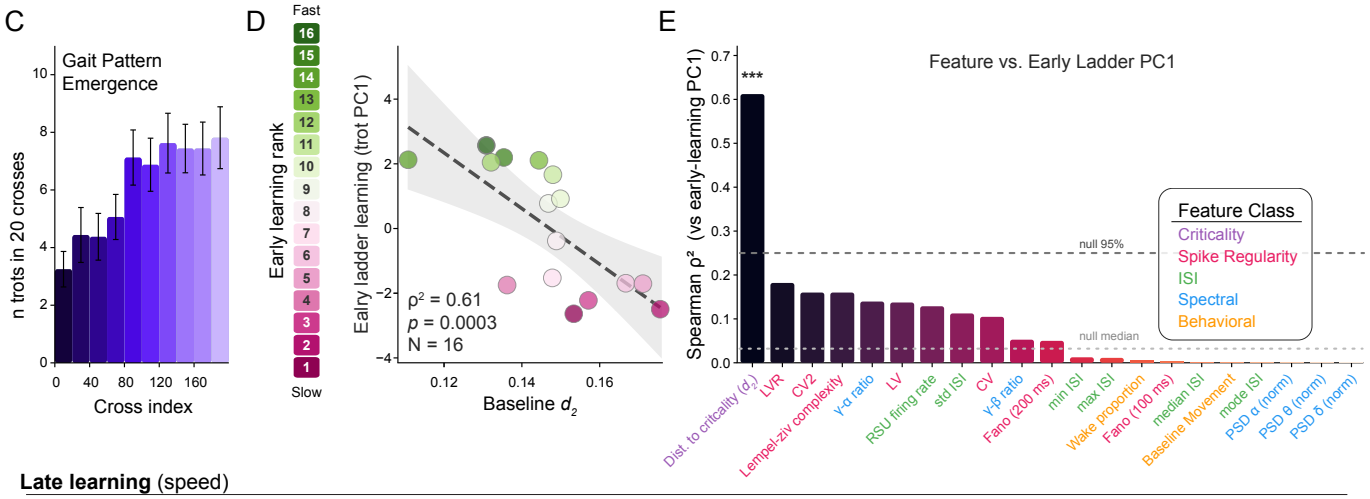
In parallel with learning to trot, animals sped up their directed ladder traversals (trots and walks) over the first 500 crossings: on-ladder duration fell in 15 of 16 animals (median Spearman $\rho = -0.34$; sign test $p = 2.6 \times 10^{-4}$), the lone exception ($\rho = +0.17$) being among the quickest at the outset (Fig. 2C). Considered over thousands of crosses, animals approached their own speed asymptotes. In principle, this latter phase might be independent of proximity to criticality, as it does not require exploration of a solution space, only the rapid execution of a previously learned motor pattern. We tested this by evaluating how well baseline proximity to criticality predicted a three-metric composite of cross speed, considered from all crosses generated in the six day protocol (Fig. S7D-F; note that there was neither a correlation between total crosses and d_2 nor the number of crosses required to reach top speeds). In contrast to our hypothesis, baseline proximity to criticality accounted for 31% of variance ($\rho^2 = 0.31$) in late learning as indicated by fast crosses ($p_{\text{perm}} = 0.027$, $N = 16$; Fig. . 2G), albeit with a lower prediction strength than early learning (late and early learning are correlated ($\rho = +0.60, p = 0.014$)). As in early learning, distance from criticality was the only baseline measure that predicted outcomes beyond chance (Fig. 2H). While below chance, the relative ordering of the 20 additional features was different from their rank in early learning. As with prey capture, asymptotic performance for both trot learning and cross speed refinement were not predicted by d_2 (Fig. S6C,D). Taken together, these data suggest that a network's basal proximity to criticality establishes learning rate not only for tasks that are variable, integrative, and evolving, but also for those characterized by stereotyped patterning and refinement[52,53]. In simple terms, a brain's proximity to criticality appears to control both the ease with which solutions can be found as well as the pace at which they are refined on repetition.

Networks at criticality have the widest possible search space while retaining the ability to stabilize (i.e., exploit) favorable patterns[7,54]. Because the motor cortex directly contributes to and constrains an animal's behavior, we reasoned that the same phenomena should be evident in the geometric structure of behavioral exploration during learning. To test this, we analyzed each animal's first 200 ladder interactions and calculated the geometric distance between crosses (MDS[55–57];

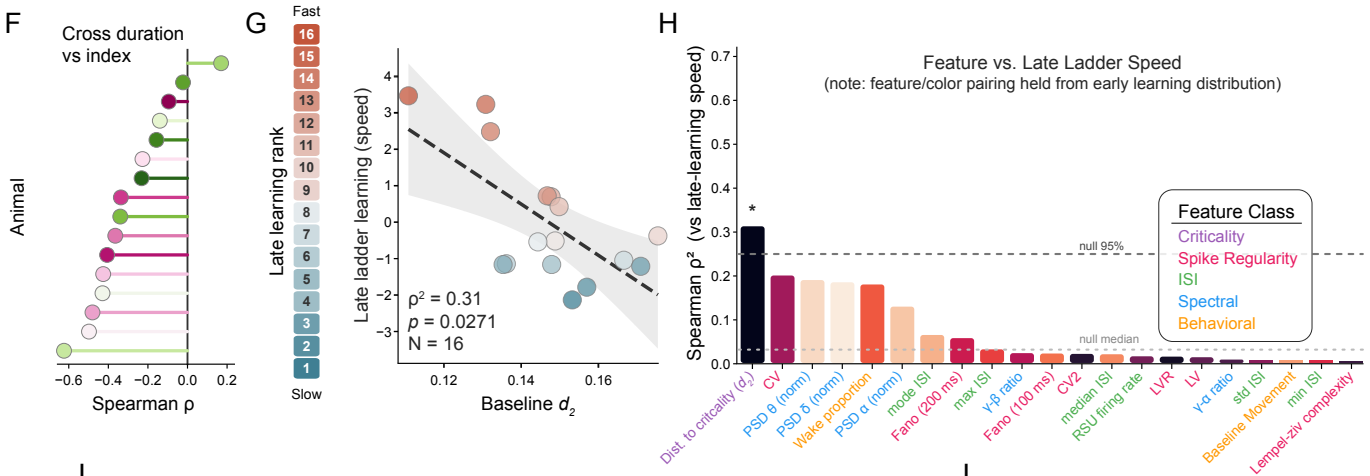
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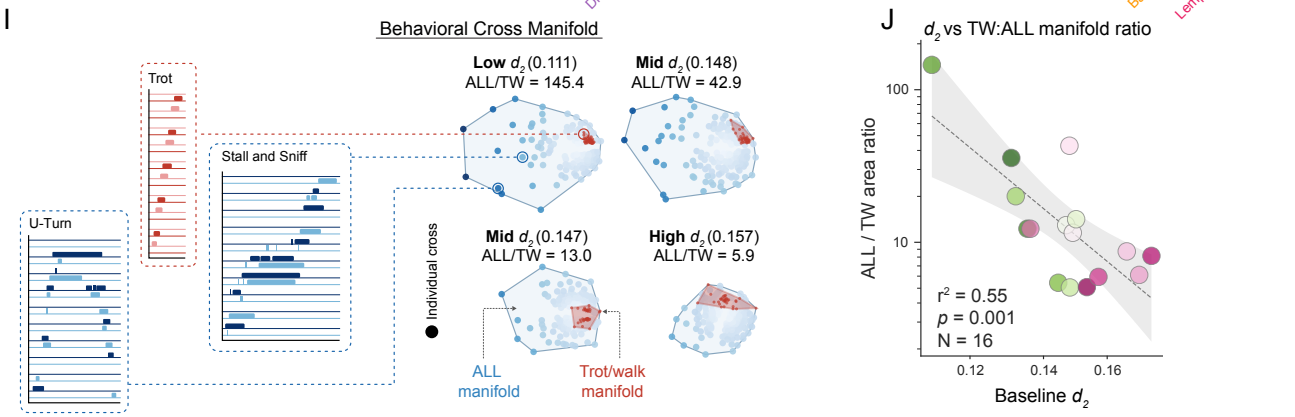


Early learning (trot patterning)



Late learning (speed)





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Figure 2: Proximity to criticality predicts ladder-cross learning rate. Across wild-type mice ($N = 16$), an individual's baseline distance from criticality in M1 predicts how it discovers and then refines a stereotyped motor sequence: ladder crossing. **(A)** Experimental design: M1 single-unit ensembles were recorded across two baseline days and used to predict learning over a subsequent six-day ladder-crossing protocol. **(B)** Representative rung-contact time series (capacitive touch sensors; left rungs, navy blue; right rungs, teal) for three cross types: an early unskilled cross, a U-turn, and a stereotyped trot. **(C–E)** *Early learning (trot patterning).* **(C)** Gait-pattern emergence: the number of trots per 20 crosses rises with cumulative ladder experience in the first 200 crossings. **(D)** Baseline d_2 predicts the rate of trot acquisition, quantified as a four-measure composite (trot PC1; $\rho^2 = 0.61$, $p_{\text{perm}} = 0.0003$, $N = 16$; dashed line, fit; shaded band, 95% CI); individuals are colored by early-learning rank. **(E)** Twenty one baseline neural, spectral, and behavioral features were evaluated for the capacity to predict the future learning of trotting across the ladder. Only d_2 cleared the 95th-percentile permutation null (dashed lines, null median and 95th percentile; asterisks). **(F–H)** *Late learning (speed).* **(F)** Animals accelerate crossing speed with experience: across each animal's first 500 purposeful (trot/walk) crossings, the within-animal Spearman correlation between on-ladder duration and crossing index is negative in 15 of 16 mice (median $\rho = -0.34$). **(G)** Baseline d_2 predicts the rate of speed refinement, quantified as a three-measure cross-duration composite ($\rho^2 = 0.31$, $p_{\text{perm}} = 0.0271$, $N = 16$); individuals are colored by late-learning rank. **(H)** Among the same features, d_2 remains the only predictor above the 95th-percentile null; feature identity and color are held from panel E to illustrate that the ordering of metrics reshuffles between early and late learning. **(I–J)** *Behavioral exploration and consolidation.* **(I)** Ladder crosses embedded by multidimensional scaling (MDS) of their touch sequences; the convex hull of an animal's first 200 crosses (ALL manifold, blue) contains the trot/walk subspace (red). Four animals spanning the d_2 range show that animals closer to criticality (lower d_2) pair a larger exploratory ALL manifold with a tightly consolidated trot/walk core (larger ALL/TW ratio). **(J)** The ALL/trot-walk area ratio is negatively correlated with baseline d_2 (Pearson $r^2 = 0.55$, $p = 0.001$, $N = 16$; log y-axis; power-law fit selected by BIC): animals closer to criticality combine the broadest exploration with the most consolidated purposeful core.

Fig. 2I). The complete set of 200 crosses comprised an *All* manifold whose area indicates the total breadth of ladder behaviors. A subspace, the *Trot/Walk* manifold, captures the regularity of purposeful crossings. These manifolds are analogous to the explore—exploit dilemma[58,59]. Measured in M1, increasing proximity to criticality suggests a larger possible repertoire of neuronal patterns. In perfect harmony, M1 baseline d_2 was negatively correlated with the All manifold area ($\rho^2 = 0.32$, $p = 0.025$; Fig. S8C); animals whose motor cortices were closer to criticality explored a larger range of ladder behaviors. Simultaneously, baseline d_2 was *positively* correlated with trot/walk manifold area ($\rho^2 = 0.25$, $p = 0.047$; Fig. S8D), suggesting that more critical animals identified and consolidated a solution set more rapidly; representative embeddings spanning the d_2 range are shown in Fig. 2I. Considered together, these findings suggest that proximity to criticality offers a mechanism linking exploration and exploitation. Consistent with this, the ratio of All to Trot/Walk manifolds was negatively correlated with baseline d_2 (Fig. 2J; Pearson $r^2 = 0.55$, $p = 0.001$ on log–log axes, power-law fit selected by BIC because the All/Trot-Walk ratio is an unbounded, magnitude-meaningful quantity): animals closer to criticality paired the broadest exploratory repertoire with the most tightly consolidated purposeful core.

Taken together, the mouse data suggest that proximity to criticality is a stable feature of network dynamics that distinguishes animals and constrains subsequent learning. The two tasks were chosen to be maximally dissimilar, potentially indicating a general principle for how neural dynamics constrain learning. However, an equally plausible alternative explanation is that d_2 captures a latent variable related to task engagement necessary for an animal to demonstrate improved performance but distinct from the mechanisms of learning.

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Proximity to criticality predicts plasticity of single neuron tuning

Ultimately, proximity to criticality should determine the ability of a complex network to generate the correlations that drive associative plasticity, whether in synthetic or biological systems. To directly test this, and sidestep confounding factors such as attention and motivation, we capitalized on the long-understood role of patterned visual experience in shaping isocortical responses[61].

In ferret primary visual cortex (V1), single neuron direction selectivity emerges in an experience dependent, experimentally controllable fashion[62] (Fig 3 A). According to our hypotheses, the magnitude of plasticity that stimuli can drive should be regulated by proximity to criticality. To test this, we combined previously published data[60] ($N = 5$) with ten new recordings from ferret kits. Briefly, to isolate stimulus-driven plasticity (eliminating variability in behavior and brain state), kits were lightly anesthetized while the activity of V1 neurons was recorded. Kits were then exposed to drifting gratings for multiple hours, after which increases in direction selectivity were robustly evident, albeit to varying degrees across animals (Fig 3 B,C; slope $= +0.0095$ $(1-\text{DirCV})\,\text{h}^{-1}$, $p = 3.0 \times 10^{-7}$; linear mixed effects model: direction selectivity $\sim$ exposure time $+ (1\,|\,\text{animal})$, where animal is included as a random effect; estimated marginal means of 0.12 and 0.21 at 0 h and 9 h, respectively; $n = 12$ motion animals).

We calculated d_2 from ensembles of V1 units prior to training blocks, allowing us to regress initial conditions against the degree of subsequent plastic change in the tuning of those same neurons. Notably, and consistent with M1 dynamics in freely behaving mice, baseline d_2 in kit V1 exhibited a range of values across animals. Individual pre-stimulus-block measurements spanned 0.068–0.187; per-animal means 0.087–0.150, a 1.7-fold range. Thus, even in the context of uniform anesthetic application, animals populated a continuum of proximities to criticality that was wider than the spread of spike-shuffled control d_2 (2.4-fold greater variance; Levene's test, $p = 0.037$) and that served as the predictor axis for subsequent plasticity.

Directly contradicting our alternate hypothesis — that d_2 might reflect some aspect of desire or engagement — pre-stimulus d_2 in V1 was significantly negatively correlated with the pending change in direction selectivity (Fig 3D; linear mixed-effects model, plasticity $\sim d_2 + (1\,|\,\text{animal})$: $\beta = -0.86$, $p = 1.1 \times 10^{-9}$; $r^2 = 0.54$ across $n = 33$ blocks from $N = 14$ kits; exposure block did not modify the relationship). The effect held when collapsed to one independent point per animal ($N = 14$: Pearson $r = -0.66$, $r^2 = 0.43$, $p = 0.010$). Conceptually, these results directly parallel the mouse ladder and prey capture data; across species, tasks, and developmental stages, animals closer to criticality exhibited more experience-dependent change. Further, of the 20 additional baseline factors, 13 were relevant to the anesthetized kits (behavior and spectral features were excluded, as animals were not behaving and we only had spike times from previous work). Consistent with both mouse behavior paradigms, proximity to criticality far outperformed the second highest ranked feature (Fig 3E). In this case, that was lempel-ziv complexity, which ranged from fourth to the least predictive feature across our other experiments. d_2, in contrast, was the primary feature in every single context.

Taken together, our data are consistent with the hypothesis that proximity to criticality mechanistically determines plasticity rate. However, in complex, interacting systems, mechanism is challenging to define and address while avoiding mereological fallacy[63]; the *phenomenon* defines the level at which the mechanism operates[64,65]. For our purposes, the phenomenon is learning capacity, and the mechanism that explains it operates at the level of population dynamics: proximity to criticality appears to constitute the organized dynamical condition under which maximal information processing — and thus maximal plasticity — is possible. This level of organization is independent of lower details, thus the mechanism should hold equally in synthetic as in biological

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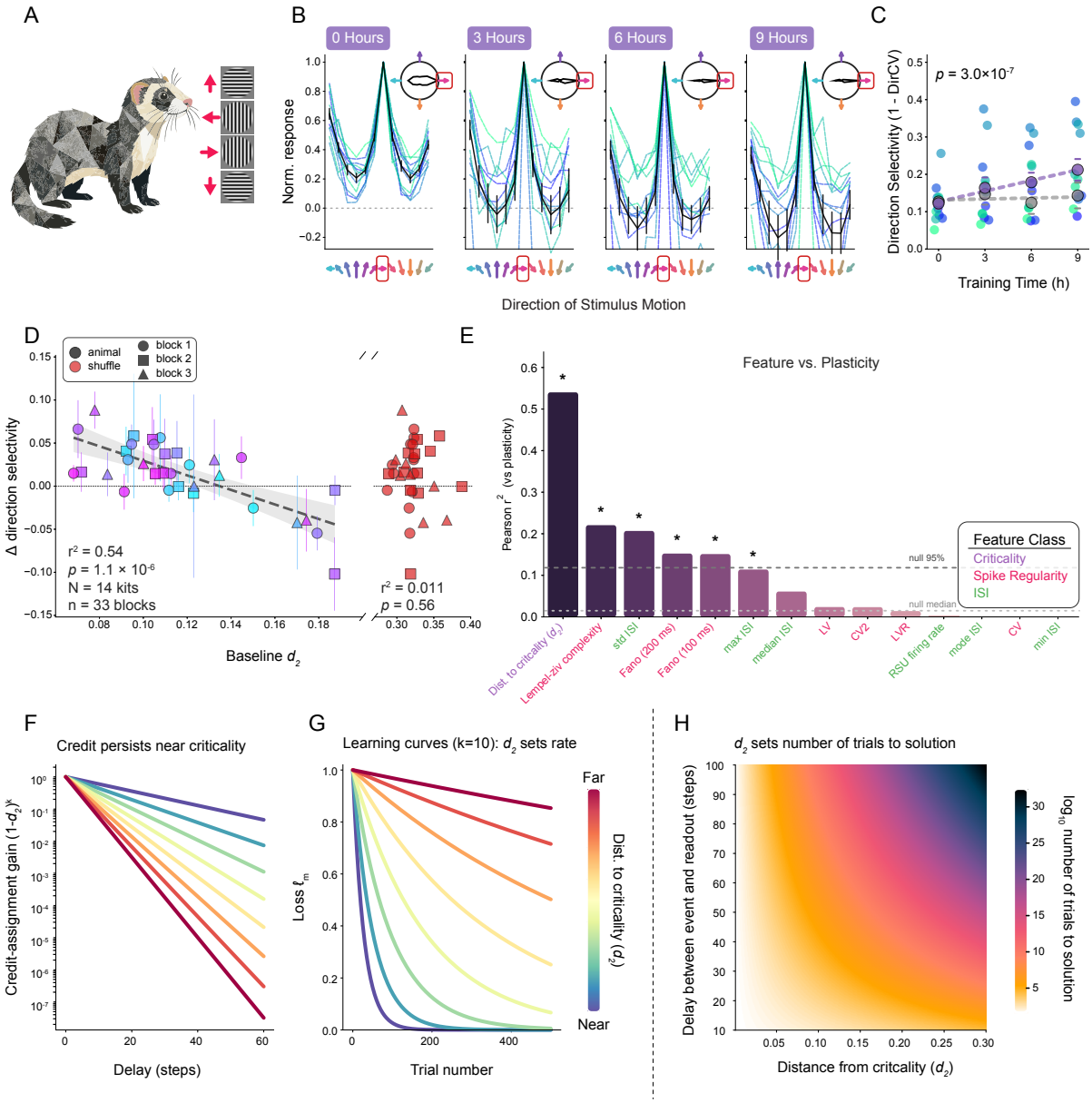


Figure 3: Proximity to criticality predicts developmental plasticity in ferret kits and defines learning rate in a synthetic learning model. A-C: Naive ferret kits ($N = 15$) exposed to moving stimuli **(A)** over hours **(B)** exhibited rapid increases in direction selectivity (individual lines represent animal-level means) **(C)** in ferret visual cortex[60]. **(D-E)**: Proximity to criticality predicts the degree of plasticity of V1 units per animal. Across a total of 33 blocks that met inclusion criteria, the degree of change in direction selectivity for units measured in that block was predicted by baseline d_2 for V1 while no correlation was observed in the shuffle control **(D)**. Compared to 13 comparator features available in the ferret dataset, baseline d_2 was the strongest predictor of the degree of plasticity in direction selectivity. **(F-H)**: A simple recurrent neural model with back-propagation reveals that proximity to criticality defines the speed for learning a task. The credit-assignment penalty associated with delay from input is shallow when d_2 is low but steepens markedly as d_2 increases **(F)**. For a fixed delay (k=10), the number of trials required to minimize loss to a minimal tolerance increases markedly with distance from criticality **(G)**. When the trials-to-solution surface is considered for a range of delays (k) and across a spectrum of d_2 values, the number of trials to the solution increases super-exponentially **(H)**.

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learning systems. Synthetic learning systems afford the opportunity to manipulate distance from criticality independently of the task and the learning rule.

Proximity to criticality constrains learning rate

In the present work, we use learning rate in the behavioral sense: the rate at which task performance improves with experience. This is different from the commonly known gradient descent step size α used below, which is a parameter controlling the size of each weight update. To ask whether the observed relationship between proximity to criticality and learning rate is a generic property of tunable information-processing systems rather than a specific principle by which d_2 constrains learning rate, we turned to a minimal recurrent model for learning (Fig. 3F–H). Mathematically, even low-dimensional dynamical systems can exhibit criticality when approaching a bifurcation[31]. At a critical bifurcation, such systems exhibit critical slowing down, i.e., autocorrelation time diverges. In simple terms, the memory of past events persists longer the closer the system is to a bifurcation[66]. In a simple recurrent neural network trained by gradient descent, this same timescale sets the limit on temporal credit assignment: for an event I that happened k time steps in the past, the present error signal is governed by the product of recurrent dynamics, reducing to μ^k (product of Jacobians), such that it vanishes ($\mu < 1$) or explodes ($\mu > 1$) unless μ is 1.0[67,68]. Over long timescales, then, we can reduce the RNN to its dominant real mode which gives a scalar recursion $h_t = \mu\, h_{t-1} + I_t$. This is the definition of an AR(1) process for which the same distance-to-criticality we measure in neural data reduces to $d_2 = 1 - \mu$ with $\mu \in (0, 1)$; the model's distance from its bifurcation is measured directly by d_2. For a linear readout learning a delayed association by gradient descent under squared-error loss, the number of trials needed to reach a fixed error tolerance ϵ (with asymptote $\ell^* = 0$) is

$$M(d_2, k) \;=\; \frac{\ln(\epsilon/\ell_0)}{2\,\ln\!\left[\,1 - \alpha I^2 (1-d_2)^{2k}\right]} \;\approx\; \frac{\ln(\ell_0/\epsilon)}{2\,\alpha I^2\,(1-d_2)^{2k}} \;\propto\; (1-d_2)^{-2k}, \tag{1}$$

where ℓ_0 is the initial loss, α the gradient descent step size, I the input drive at the first time step, and k the temporal horizon over which credit must be assigned (see Methods).

This one-parameter model reproduces the three principles that organize the animal data. First, it makes explicit the mechanism by which proximity to criticality matters: the credit-assignment gain $(1 - d_2)^k$ — the fraction of an error signal that still reaches an event k steps in the past — decays geometrically with delay, with a decay constant fixed entirely by d_2 (Fig. 3F). Near criticality the gain persists across long delays, so that temporally distant events remain visible to learning; away from criticality the gradient vanishes within a few steps, and the network is effectively blind to its own recent history. Second, feeding this surviving gradient into gradient descent determines learning *rate*, not asymptote, validating the principle that the speed of learning (but not its ultimate outcome) depends on d_2. Every network can in principle reach zero loss, but the speed of that descent scales with $(1 - d_2)^{-2k}$: near-critical networks converge in tens of trials while more distant ones require hundreds to thousands, all settling onto the same floor (Fig. 3G). This is precisely the dissociation observed in the animals — baseline d_2 forecasts how quickly a mouse improves and how much a kit's tuning changes, but not the endpoint it eventually reaches. Third, because the cost scales as $(1 - d_2)^{-2k}$, the penalty for being off-critical compounds with task difficulty. The trials-to-solution surface (Fig. 3H) shows that for short credit horizons, slightly non-critical networks can learn in a feasible number of trials[13], but as k grows the required trials rise exponentially with distance from criticality.

That a one-parameter, generic recurrent network reproduces the same d_2–learning relationship

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seen across two mouse tasks and ferret visual plasticity indicates that the causal link between baseline proximity to criticality and the capacity for plastic change is a generic property of learning in recurrent systems operating near criticality.

Proximity to criticality predicts general intelligence in humans

Much of human intelligence is necessarily acquired: one cannot speak a language without having been exposed to it and having undergone durable, experience-dependent change to retain it. Such intelligence is, by definition, a product of plasticity. This distinguishes it from capacities such as the stretch reflex, which is present in neonates and requires no experience to elicit[69]. Our animal and modeling data raise the possibility that, if high-resolution cortical electrophysiology were available in relevant human circuitry, proximity to criticality might provide a mechanistic explanation of durable, inter-individual variation in cognitive performance. Because human validation of our hypothesis requires both high-resolution neural data and normed cognitive assessment, analysis necessitates a clinical cohort. With this in mind, we examined a stereoelectrode-based electrophysiology dataset collected from 35 patients undergoing chronic monitoring in a children's epilepsy clinic (Fig. 4A). As d_2 can be measured in as little as 1,000 samples[40], we curated a relatively small sample ($\sim$ 60 min) of artifact free data from a frontal isocortical probe in each patient during waking. Note that, proximity to criticality in EEG is indistinguishable between epilepsy patients and healthy controls when taken from inter-ictal periods removed from seizure activity[70]. In our dataset, each patient's full scale intelligence quotient (IQ) was assessed by clinicians. While IQ collapses heterogeneous abilities into a single score and has a history of misuse[71], it reliably and predictively indexes a durable, high-level capacity for adaptive cognition: full-scale scores are among the most stable and reproducible measures in all of psychology, with high test–retest reliability, rank-order stability that persists across decades[72], and predictive validity for educational, occupational, and health outcomes that exceeds that of any comparable single variable[71,72].

Importantly, IQ tests probe the ability to solve novel problems, not previously acquired knowledge or skills. Thus, they aim to quantify the capacity to acquire and integrate, which is, conceptually, the plastic operation our data suggest arises from proximity to criticality. To test this directly, we computed d_2 from frontal stereo EEG. Because clinical stereo EEG resolves mesoscale local field potentials rather than single units, we fit the autoregressive model directly to continuous data; the high-gamma envelope serves as a proxy for local multi-unit bursting[75,76], and its long-range temporal correlations are preserved beyond the recording hardware's high-pass cutoff (Fig 4B). In the patient cohort, frontal cortical proximity to criticality correlated with full-scale IQ: lower d_2 was associated with higher IQ after adjusting for demographic covariates (age, sex, race/ethnicity, and handedness; partial $r = -0.60$, $r^2 = 0.36$, $p = 3.8 \times 10^{-4}$, $N = 35$; Fig 4C,D), an effect that was already significant at the zero-order level (Pearson $r = -0.51$, $r^2 = 0.26$, $p = 1.7 \times 10^{-3}$). In contrast, in our cohort of epileptic patients, full-scale IQ did not vary significantly with seizure burden (Spearman $\rho = -0.12$, $p = 0.49$, $N = 35$). Similarly, although in this cohort IQ increased with age across the 2–22 y range ($r = +0.39$), d_2 was uncorrelated with age ($r = -0.17$, $p = 0.34$), and differed neither by sex (point-biserial $r = +0.01$, $p = 0.94$; Cohen's $d = 0.03$), nor race/ethnicity (point-biserial $r = +0.01$, $p = 0.97$; Mann-Whitney $p = 1.00$).

As in the animal experiments, we asked whether this prediction generalized to other features of the signal. Because sEEG resolves mesoscale field potentials rather than spikes - and because absolute signal amplitude is not comparable across electrodes - we competed d_2 against 13 amplitude-invariant spectral features. Of the set, only d_2 predicted IQ above chance (next-best feature $r^2 = 0.10$, $p_{\text{perm}} = 0.07$; Fig 4E), mirroring its dominance over the metrics evaluated in mice and ferrets.

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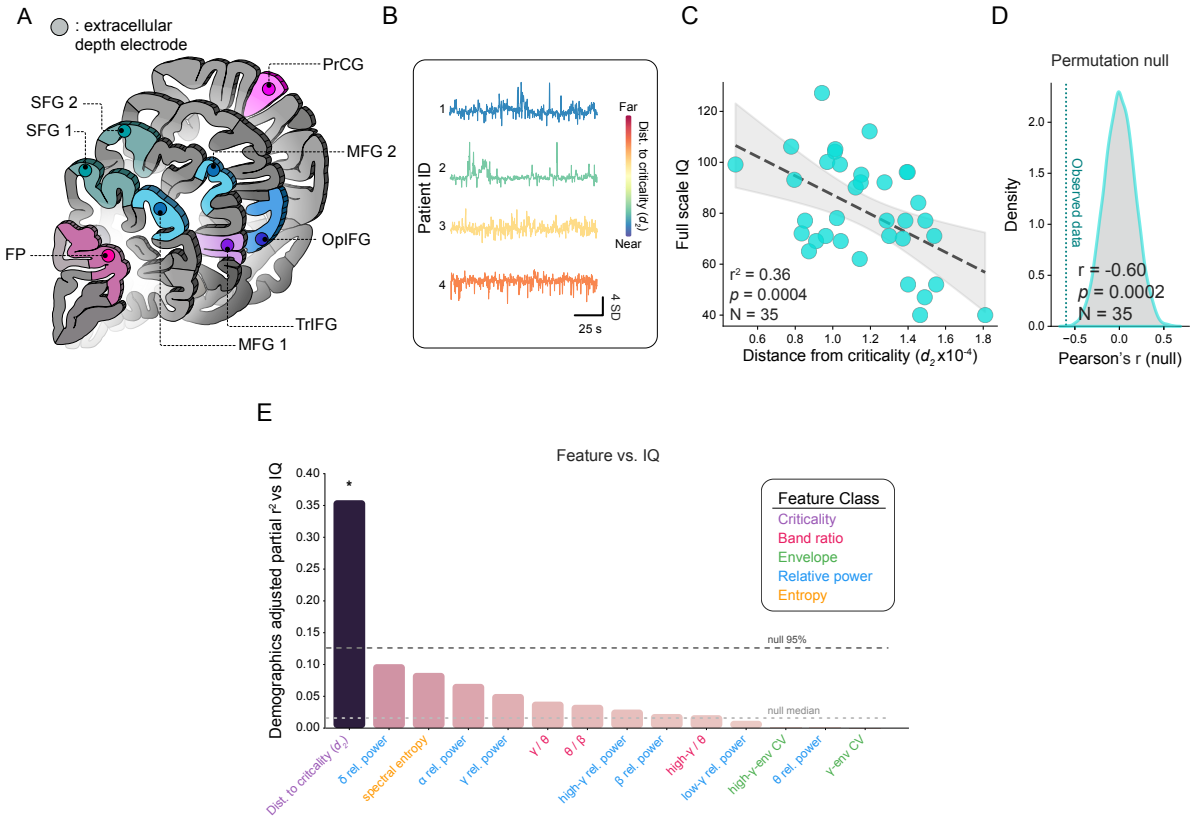


Figure 4: Proximity to criticality predicts general intelligence in human patients. **A**: Frontal isocortical entry sites of the analyzed stereoelectrode rods, shown on representative coronal plates from the Allen Human Brain Atlas[73,74] (Allen Institute for Brain Science; `human.brain-map.org`). Each marker denotes the superficial cortical entry of one patient's analyzed rod and indicates regional coverage rather than an exact contact coordinate. Note that one brain region (PrCG, $N = 2$ patients) is not represented as this region is rostral to the first available Allen Human Brain Atlas plate. For another patient ($N = 1$) the region is unassigned, though confirmed to be in the frontal cortex **(B)**: Exemplar enveloped stereo EEG recordings show multiple timescales that vary with proximity to criticality. **(C-D)**: Isocortical proximity to criticality predicts IQ in a clinical epilepsy population ($N = 35$) **(C)** and a permutation test confirms that this relationship could not have arisen by chance **(D)**. **(E)**: Comparison of d_2 against comparator features reveals that proximity to criticality is the only neural feature able to predict IQ in our dataset.

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Previous work has shown an association between proximity to criticality and general intelligence in adults and children, based on functional imaging or surface EEG-based measures[77–79]. Our results extend this literature by using direct intracranial recordings alongside a time-domain criticality measure, and we found a strong predictive relationship. When considered alongside the mouse, ferret, and model data, our findings argue that isocortical proximity to criticality constrains a common capacity for experience-dependent change, conserved from mouse to human.

Discussion

The brain is the source of the adaptive, complex behavior required to thrive in an unpredictable world. Because behavior is the substrate for evolutionary selection[80], animals that adapt more quickly and effectively should out-compete others. As a result, brains, regardless of anatomical specifics, might be expected to ultimately converge on the computational principles that maximize adaptability[10]. Through this lens, due to its direct, *a priori* explanatory power regarding behavior and function, criticality has been suggested as a plausible unifying feature of brain function across species and an endpoint of homeostatic plasticity[22,81]. Consistent with this, proximity to criticality degrades in pathologic states that compromise adaptability[82]. What has remained untested, however, is the central prediction that this endpoint constrains adaptive plasticity in the brain. Here, we provide multiple lines of evidence supporting that conclusion. We find that, just as with other homeostatic set-points in physiology[41,42,83], in vivo brain networks exhibit a range of near-critical set-points that are stably maintained at the level of the individual. Animal-level variation within this range appears to consequentially determine the rate at which an individual can drive plastic change, such as in learning, in the future. In complex systems not amenable to interventionist tests, such as economies[84] or ecosystems[85], consistent predictive power is a test of causality[86]. To go beyond this, we establish causality using a synthetic learning model. Finally, we present evidence for the existence of a similar relationship in the adaptability of the human brain.

Our findings also speak to the function of sleep. Sleep is required for adaptation across the animal kingdom, from respiratory plasticity[87] to complex cognition[88] — a universality that is hard to explain if sleep services a catalog of circuit-specific needs. Occam's razor favors a single shared variable, necessary for adaptive computation and restored by sleep; proximity to criticality fits, being homeostatically maintained[22], eroded across waking and renewed by sleep[23,89], and progressively, sleep-dependently lost in neurodegeneration[82]. Tellingly, although innumerable features differ between sleep and wake, it is distance from criticality — not slow-wave energy or any other measure — that predicts impending state transitions[23]. Our results demonstrate that the same set-point governs the rate of future plastic change, the very capacity that fails without sleep.

The same logic links dynamics and learning. Prior work has tied spectral features[90,91], neural complexity[92], and spike-time irregularity[93] to learning, plasticity, or cognitive ability. Our analytical and empirical results show these are subsumed by proximity to criticality. Because d_2 is a geometric property of the population's dynamics, many such features must covary with it. The covariance of d_2 and other features tends to be local, as the relationships shift across the space (Fig. S2), such that none reconstructs d_2 and d_2 cannot be trivially built from any of them. However, d_2 does not exist in a vacuum: autocorrelation time offers a familiar though not independent bridge to distance from criticality[94]. Autocorrelation time and susceptibility, like long-range temporal correlations[34,95] and the branching ratio[96,97], are signatures of an RG fixed point, each diverging or becoming singular as the system nears one (critical slowing down). Each offers an indirect readout of proximity to criticality, whereas d_2 measures that distance directly in the space of AR models. As a confirmation, a panel of autocorrelation-based statistics (integral and timescale, AR-dominant timescale, and susceptibility) also predicted learning across all three tasks, and their

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theoretical dependence on the AR coefficients tracks the critical boundary but less directly and less uniformly than d_2 (Fig. S9). Because they are indirect, their contribution was variable within and across tasks: d_2 remained the strongest predictor for both trot-emergence and prey-capture learning, and was surpassed only in late-phase ladder refinement (speed) — the setting in which we had predicted the smallest role for criticality and observed the weakest d_2 effect.

The deepest implication of our results is not specific to a task, region, or species, but the relationship between dynamical organization and computation itself. It is increasingly clear that a system's capacity to learn is constrained by mathematical principles that are largely indifferent to its physical substrate; learning arises in cephalopods[98] as it does in LLMs. The same distance-to-criticality that governs credit assignment in our recurrent model appears to govern learning rate in three unique mammalian brains. This advances the hypothesis[11,50] that proximity to criticality is not an abstract correlate of cognition but constitutes the dynamical regime that accounts for the adaptive computation underlying much of behavior and brain function. If so, the principle should extend to any aspect of cognition that depends on plastic change. We emphasize, however, the boundaries of what our data establish. The relationship is predictive across species and causal within a model; a direct demonstration that manipulating proximity to criticality alters learning rate in the brain remains untested. Further, the human association is cross-sectional, and whether the same principle holds in nervous systems that evolved independently of the mammalian cortex is unknown, although prior work describes near-critical dynamics in, e.g., leeches, fish, and turtles[10].

Proximity to criticality thus offers a mechanistic account of variability in learning rate at the level of population dynamics[65], and represents a theory-derived prediction[1,4,99] borne out across species, regions, and paradigms. It does not specify the lower-level machinery that establishes the set-point — synaptic homeostasis[100], cell-non-autonomous network governance[101], and the sleep-dependent restoration described above[23] are candidates. However, the theory and growing empirical evidence suggest that diverse underlying machinery establishes a unifying, emergent computational regime necessary for experience-dependent plastic change. From this perspective, each cellular and molecular component bears on learning only insofar as it moves the brain toward or away from criticality, or establishes the capacity for plastic modulation of connection weights (e.g., input-specific associative plasticity). In simple terms, proximity to criticality may reflect the speed limit of adaptation in complex learning systems, whether biological or artificial.

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Methods

Mice

All procedures were performed in accordance with protocols approved by the Washington University in Saint Louis Institutional Animal Care and Use Committee, following NIH guidelines for the Care and Use of Laboratory Animals. In total, 12 female and 6 male adult wild-type C57BL/6 mice were used (Jackson Labs, strain #000664). Mice were postnatal day 65 (P65) to P123 at the onset of baseline recording (mean = 80.0 d, s.d. = 15.9 d). All mice were housed in an enriched environment on a 12:12 h light/dark cycle.

Surgery. Custom-built multielectrode arrays were assembled and implanted as described previously[23,39,82]. Prior to surgery, mice were administered buprenorphine SR (0.1 mg kg^{-1}), dexamethasone (0.5 mg kg^{-1}), and meloxicam (5 mg kg^{-1}). Mice were anesthetized with isoflurane (3% for induction, 1–2% for maintenance, in air) and head-fixed in a robotic stereotaxic instrument. The fur, skin, and periosteum over the dorsal skull were removed, and a 1–1.5 mm craniotomy was cut over the target region. Custom-built, 64-channel, tetrode-based arrays were lowered into each target at 5 mm min^{-1} using a robot and vacuum-based probe holder. All 18 mice received a 64-channel array in the right M1 caudal forelimb area (coordinates, AP/ML/DV relative to bregma and dura: 0.25/1.25/0.79 mm). All animals were implanted in the right hemisphere so that hemisphere was held constant across the cohort, removing inter-hemispheric variability as a confound in between-animal comparisons of d_2; the right hemisphere was chosen arbitrarily. In 5 of the 18 mice, an additional, second 64-channel array was implanted in right primary visual cortex (V1; coordinates −3.20/2.20/1.00 mm), targeted as in Parks et al.[39], yielding 128 channels across the two regions. Unless otherwise noted, analyses reported here use the M1 ensembles. A bridged ground/reference wire was connected to a skull screw over the contralateral hemisphere; the array was connected to headstage electronics (White Matter LLC) via custom PCB-EIBs. Electronics were enclosed in a 3D-printed case, and secured with dental cement. The entire assembly weighed $< 3\,g$. Mice received meloxicam (5 mg kg^{-1}) and dexamethasone (0.5 mg kg^{-1}) for 3 d post-operatively and recovered for a minimum of 7 d before recording began.

Recording. Recordings were performed continuously in freely behaving mice across three environments: a standard home cage during the baseline period, a circular prey-capture arena, and a custom ladder cage (both described below). Throughout recording, mice were connected via a custom cable containing an in-line commutator (slip ring) that permitted unrestricted movement. Neural signals were amplified, digitized (14-bit), and acquired at 25 kHz on the eCube acquisition system (White Matter LLC). Behavioral video was captured at 15 or 30 frames s^{-1} and temporally aligned to the neural data. For each animal, a continuous 13–14 d recording (312–336 h) comprised a 24–48 h baseline of spontaneous home-cage behavior followed by two 6-day motor-learning tasks; task order (prey capture first vs. ladder first) was randomized across animals.

Prey-capture task. Prey were young-adult Turkestan red-runner cockroaches (*Shelfordella lateralis*; Caribbean Mealworms) weighing 0.05–0.20 g (mean = 0.145 g, s.d. = 0.041 g). For prey-capture experiments, mice were housed individually in circular hunting arenas constructed from 19 L (5-gallon) buckets, which served as home cages for the 6 d protocol. On Day 1 (pre-trial), a single cockroach was introduced at 17:00 and left overnight; any cockroach not captured was removed the following morning at 09:00. On Day 2, food (but not water) was removed at 17:00, beginning a 16 h food-deprivation period. Beginning at 09:00 on Day 3, cockroaches were presented one per trial, up to six trials per day. Each trial lasted < 30 min: if a mouse did not engage or failed to hunt successfully, the cockroach was replaced. If a mouse captured its prey rapidly, a 10 min delay was introduced before the next cockroach was presented. This 16 h cycle of food deprivation

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and hunting was repeated on four consecutive days; each mouse encountered 25 cockroaches in total (4 days × 6 trials, plus the pre-trial cockroach). Mice had *ad libitum* access to food from Day 1 through the evening of Day 2; water was available *ad libitum* at all times except during the four hunting sessions.

Automated ladder-rung walking task. To embed a structured motor challenge within the home cage, we developed a self-paced ladder paradigm in which a horizontal ladder separates the food and water sources, so that the mouse must traverse it to feed and drink (Fig. 2B). The ladder was made from laser-cut acrylic and T-slotted aluminum extrusion (80/20). Mice were kept on a 12:12 h light/dark cycle (visible lights on at 07:00, off at 19:00), with infrared lights on continuously so cameras could image animals throughout the dark phase. Each ladder cross constituted a single "trial," yielding a naturally repeating trial structure while preserving spontaneous, ethological behavior. Over days, mice generated thousands of crosses spanning the full course of learning, from initial acquisition through a stabilized motor solution.

Behavior was captured from both sides of the ladder by two Raspberry Pi High Quality (HQ) cameras (12 MP) recording at 60 fps, each driven by a dedicated Raspberry Pi 4B. Rung contacts were sensed by two MPR121 12-channel capacitive touch controllers (20 channels in use, encoding 10 left–right rung pairs) sampled at 1 kHz and read by a single Raspberry Pi 4B. To permit simultaneous neural recording, the capacitive sensors were electrically isolated from the controlling Raspberry Pi with bidirectional, I^2C-compatible digital isolators (ISO1640, Texas Instruments), preventing bus noise from contaminating the neural recording. All data streams—video-based limb kinematics and rung-contact times—were synchronized to the electrophysiological recording. Inspired by the Precision Time Protocol[102], a Raspberry Pi Pico broadcast a synchronization pulse to all recording devices every 10 s, and the streams were aligned post hoc to this shared reference clock.

Spike sorting and unit classification. Only baseline-period neural data were analyzed in this study. Broadband data were band-pass filtered between 500 and 7,500 Hz. Candidate spikes were detected by a threshold between 4.5 and 6 standard deviations below the mean of the filtered signal; for each animal, the threshold yielding the greatest number of single units was selected. Spikes were sorted with a modified implementation of MountainSort5[103] and SpikeInterface[104] in non-overlapping 4 hour clustering blocks. Clusters were classified by an XGBoost model using features including waveform shape, waveform consistency, inter-spike-interval distribution, and multi-channel spatial spread[105]. Cluster quality was confirmed manually by trained scorers blinded to behavioral outcome. Each unit was assigned to one of four quality tiers; all analyses were restricted to well-isolated single units (Quality 1 and 2), excluding multi-unit (Quality 3) and noise-contaminated (Quality 4) clusters. Single units were further classified as regular-spiking (RSU) or fast-spiking on the basis of spike waveform[106]. All analyses reported here include only RSUs, and blocks with $n \leq 15$ RSUs were excluded from analysis owing to inaccurate measurement of d_2 from such a limited set. Across the 18 mice, 6,326 RSUs were recorded (mean = 34 ± 13.9 per animal; range 15–71).

Arousal-state classification. Arousal state was manually scored from a combination of spectral and movement features[23,39]. For each recording, three M1 channels with clean local field potential (LFP) were downsampled to 500 Hz, and LFP power in the 0.1–60 Hz band was rendered as hourly spectrograms. Synchronized behavioral video was processed with DeepLabCut[107] for pose estimation, and the resulting movement traces were aligned to the LFP. States were assigned in 4 s epochs consistent with well-established parameters[37,108]: high delta-band power with low movement was scored as NREM sleep; low delta-band power with elevated theta-band power and no movement

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was scored as REM sleep; and elevated theta-band power accompanied by movement was scored as wake. In total, 728 h of baseline recording across the 18 mice were scored manually. Unless otherwise noted, all neural features were computed exclusively during waking epochs.

Ferrets

Subjects and general design. Ferret data were collected in the laboratory of S.D.V.H. at Brandeis University under protocols approved by the Brandeis University Institutional Animal Care and Use Committee, following NIH guidelines. The dataset comprised 15 female ferret kits (*Mustela putorius furo*), aged postnatal day 31–34 (around the time of natural eye opening, when direction selectivity begins to emerge), studied in terminal, anesthetized electrophysiology experiments. Twelve kits received visual training with drifting gratings and three received a static gray-screen control. Of the 15, five were reported previously[60] and the remaining ten were newly recorded. Female animals were used exclusively because animals were co-housed with sexually mature females and co-housing with males induces stress[60].

Surgery and anesthesia. Kits were initially sedated with ketamine and given atropine and dexamethasone to reduce secretions and inflammation. Anesthesia was induced with isoflurane in a nitrous-oxide/oxygen mixture delivered by mask, and a tracheostomy was performed for mechanical ventilation (1.5–3% isoflurane in 2:1 nitrous oxide:oxygen). An intraperitoneal cannula was placed for delivery of Ringer's solution and the neuromuscular blocker gallamine triethiodide, which suppressed spontaneous eye movements during recording. The electrocardiogram was monitored continuously to titrate anesthetic depth, and body temperature was held at 37 °C. A 4×4 mm craniotomy was made over primary visual cortex (V1) in the right hemisphere and the dura was resected.

Electrophysiology. Multiunit activity was recorded across all cortical layers of V1 with a 32-channel silicon probe (NeuroNexus, A1x32-Poly2-10mm-50-177) inserted approximately perpendicular to the cortical surface (900–1100 μm depth), with agarose applied to stabilize the brain. Signals were acquired on one of two systems (11 animals via a Multichannel Systems preamplifier/amplifier with LabVIEW and a National Instruments 6071e board; 4 animals via an Intan RHD2132 headstage). For both systems, spike waveforms were detected at a threshold of 5 standard deviations, and all threshold crossings on a channel were pooled as a multiunit cluster, because activity attributable to isolated single units was rarely resolvable with these electrodes[60].

Visual stimulation and direction-selectivity assessment. Visual stimuli were generated in MATLAB with the Psychophysics Toolbox and displayed on a Sony GDM-520 monitor 45 cm from the animal. Stimuli were full-field, high-contrast drifting sinusoidal gratings presented in 12 directions (30° increments), with a spatial frequency of 0.08 cyc deg^{-1}. The experiment alternated assessment and training blocks: direction tuning was assessed with gratings at 25 deg s^{-1} before training (0 h) and after each of three 3 h training blocks (3, 6, and 9 h), during which the animal viewed drifting gratings (motion condition) or a static gray screen (control). At each assessment, direction selectivity per recording site was quantified as $1 - \mathrm{DCV}$, where DCV is the directional circular variance[109,110]. Sites were included only if they (i) showed significant response variation across stimuli by ANOVA, (ii) responded at ≥ 1 Hz to 25 deg s^{-1} gratings, and (iii) had a preferred orientation within 30° of the optimal orientation. The whole-animal, per-block plasticity used below is the change in the cell-averaged $1 - \mathrm{DCV}$ across an exposure block.

Human stereo-EEG

Patients. We performed a retrospective, observational study of children and young adults, ages 2–25 y, who underwent clinically indicated stereotactic EEG (sEEG) monitoring to identify a

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seizure focus between February 2016 and July 2023 in the Epilepsy Monitoring Unit. The study was approved by the Washington University in St. Louis Institutional Review Board (Protocol 202412145); consent was waived as the study used observational, retrospective data. Electrodes were standard, clinically approved depth electrodes (800 μm diameter, 2 mm contact length, 3.5 mm inter-contact spacing; PMT, Chanhassen, MN). Implantation was planned per patient according to the primary hypothesis for seizure localization, with adjacent electrodes demarcating the seizure-onset zone and additional electrodes placed to identify eloquent cortex for language and motor mapping.

From a total cohort of $N = 54$ patients recorded, $N = 35$ met inclusion criteria of (1) a valid full-scale intelligence quotient (IQ) and (2) artifact-free frontal recordings during waking. These 35 patients (ages $\sim$ 2–22 y) are summarized in detail in Table 1. Full-scale IQ was determined for each patient by clinical neuropsychological assessment (WISC-V or WAIS-IV, age-appropriate version; administered by a licensed clinical neuropsychologist) as part of standard presurgical epilepsy evaluation.

Table 1: Analyzed patient demographics. Summary of the analyzed cohort ($N = 35$), each contributing a single frontal isocortical contact. Continuous variables are reported as median [IQR] with range; categorical variables as count (%). Seizure burden reflects the clinical cvEEG estimate (approximate seizures per day). ASD, autism spectrum disorder; TSC, tuberous sclerosis complex; FSIQ, full-scale IQ; IQR, interquartile range.

Characteristic	Value
Age (years)	
Median [IQR]	14 [10–18]
Range	2–22
Sex	
Male	20 (57%)
Female	15 (43%)
Handedness	
Right	26 (74%)
Left	9 (26%)
Ethnicity	
White, non-Hispanic	31 (89%)
Black/African American, non-Hispanic	3 (9%)
Asian	1 (3%)
Additional Diagnoses	
ASD	7 (20%)
TSC	6 (17%)
Prior lesion resection	5 (14%)
Seizure burden (approx. seizures/day)	
Median [IQR]	1.6 [0.5–3.0]
Range	0–20
FSIQ	
Median [IQR]	77 [70–96]
Range	40–127

Stereo EEG acquisition. Stereo-EEG uses intracranial *depth* electrodes (not scalp or subdural surface electrodes); each rod carries multiple contacts sampling local field potentials along its

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trajectory, typically on the order of 150 contacts per patient. For analysis we analyzed a single superficial electrode from each patient in a non–seizure-onset region with similar anatomic positioning across subjects (e.g., motor, frontal, or cingulate cortex). Anatomic regions were identified using CT, MRI, and Curry 3D modeling software (Compumedics, Charlotte, NC), and contact locations were reconstructed from clinically acquired imaging. EEG was acquired on a clinical Nihon Kohden system (Tokyo, Japan) using a non–DC-coupled amplifier with 250 channels, at a native sampling rate of 1,000 Hz (studies from 2016–2017 used 500 Hz); native rate varied by patient and was standardized to 500 Hz in analysis (see below). Anti-seizure medications were gradually weaned during EEG recording period. When possible, only epochs free of seizure activity were used. For some patient's with abundant seizures, epochs with the lowest seizure burden were analyzed. In all cases, seizure activity was verified by a clinical epileptologist. Consistent with prior work, inter-ictal dynamics in epilepsy patients show normal signatures of criticality[70]. Data were deidentified and exported in European Data Format (EDF); clinical and demographic data were maintained by the clinical study team.

Contact selection and preprocessing. For each patient, a single frontal isocortical contact was analyzed, and approximately 60 min of artifact-free waking data were used. Across the cohort, analyzed contacts were distributed over frontal isocortical gyri of both hemispheres (per-patient localization in Table 2; entry sites shown on Allen Human Brain Atlas coronal plates in Fig. 4A): middle frontal gyrus ($N = 12$), inferior frontal gyrus pars opercularis ($N = 12$), superior frontal gyrus ($N = 4$), inferior frontal gyrus pars orbitalis ($N = 2$) and pars triangularis ($N = 1$), precentral gyrus ($N = 2$), and frontal pole ($N = 1$); one patient carried a bilateral implant that could not be assigned to a single gyrus.

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Table 2: Localization of the analyzed frontal sEEG contact, per patient. Each patient contributed a single frontal isocortical contact ($N = 35$). Hemisphere: L, left; R, right; U, undetermined. Region abbreviations: MFG, middle frontal gyrus; SFG, superior frontal gyrus; OpIFG, inferior frontal gyrus (pars opercularis); TrIFG, inferior frontal gyrus (pars triangularis); OrIFG, inferior frontal gyrus (pars orbitalis); PrCG, precentral gyrus; FP, frontal pole. One patient carried a bilateral implant that could not be assigned to a single gyrus. Patient identifiers are the study's internal codes.

Patient	Hem.	Region	Patient	Hem.	Region
4C	L	MFG	3C	R	OpIFG
17C	R	MFG	6C	R	OpIFG
20C	L	MFG	14C	R	OpIFG
34C	L	MFG	16C	L	OpIFG
40C	L	MFG	21C	L	OpIFG
41C	L	MFG	26C	L	OpIFG
42C	L	MFG	30C	L	OpIFG
49C	R	MFG	38C	R	OpIFG
52C	R	MFG	43C	R	OpIFG
55C	L	MFG	48C	L	OpIFG
56C	R	MFG	59C	L	OpIFG
57C	L	MFG	60C	L	OpIFG
36C	L	SFG	45C	L	TrIFG
50C	L	SFG	39C	R	OrIFG
53C	R	SFG	47C	R	OrIFG
58C	L	SFG	31C	R	PrCG
			46C	L	PrCG
			23C	R	FP
			8C	U	bilateral (unassigned)

Signals were resampled to 500 Hz (polyphase resampling), notch-filtered at 60, 95, and 120 Hz (IIR notch, $Q = 30$), and high-pass filtered at 0.1 Hz (4th-order Butterworth, zero-phase). Transient artifacts were removed within each analysis window by flagging samples whose amplitude exceeded the window median by 3 robust standard deviations (1.4826× MAD), padding flagged samples by 50 ms, and linearly interpolating gaps up to 1 s; windows in which more than 10% of samples were interpolated, or which contained any longer gap, were discarded. Cleaned data were segmented into non-overlapping 180 s windows for d_2 estimation (below).

Proximity to criticality (d_2): spike-based ensembles (mice and ferrets)

Definition. Proximity to criticality was quantified as d_2, a distance-to-criticality measure derived from a renormalization-group[111] treatment of univariate time series[40]. A population time series was fit with an order-p autoregressive (AR) model, yielding coefficients $\phi_1, \ldots, \phi_p$ that specify a point in the p-dimensional space of AR(p) models. The set of perfectly critical models—those generating autocorrelations across all scales—forms a hyperplane $\sum_{\tau=1}^{p} \phi_\tau = 1$ in this space, and d_2 is the Euclidean distance from the fitted point to that hyperplane,

$$d_2 \;=\; \frac{\left| 1 - \sum_{\tau=1}^{p} \phi_\tau \right|}{\sqrt{p}}. \tag{2}$$

Throughout, $p = 10$. Smaller d_2 indicates greater proximity to criticality; white noise (all $\phi_\tau = 0$) gives a reference for 'far from criticality' $d_2 = 1/\sqrt{p} \approx 0.32$ for $p = 10$. The same definition is used for mice, ferrets, and humans; only the substrate and the estimation of the AR coefficients differ.

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Mouse d_2. Only regular-spiking, Quality-1–2 single units were included. Each baseline recording was divided into non-overlapping 5 min windows, and only windows in which the animal was awake for $> 90\%$ of the window were analyzed. Within each window, ensemble spike counts were summed into 15 ms bins to form a population spike count time series. A tenth-order AR model was fit by the Yule–Walker method with maximum-likelihood estimation (`statsmodels`), and d_2 was computed from the resulting coefficients by Eq. (2). Unless otherwise noted, each animal's baseline d_2 is the mean across its wake windows.

Ferret d_2. For ferrets, d_2 was computed per animal and exposure block from the pooled multiunit spike train, using the same distance-to-criticality definition (Eq.1) as for mice. Because ferret firing rates varied widely across recordings, the bin width DT was set per recording as a fixed multiple of the mean inter-spike interval, $\mathrm{DT} = 5 \times \overline{\mathrm{ISI}}$. Analysis was restricted to "valid" windows of sustained activity, i.e., periods with ≥ 1 spike per bin sustained for ≥ 3 s—because including inter-stimulus and blank periods allows stimulus structure to swamp the signal. The autocorrelation function was accumulated within valid windows (averaged across all valid windows, with no cross-window lag pairs), and the AR(10) coefficients were obtained from this windowed autocorrelation function by Yule–Walker method; d_2 then followed from Eq. (2). This estimator is algebraically identical to the mouse definition. Because the multi-hour anesthetized preparation can drift in depth, blocks were screened for anesthetic plane by two criteria: 1) the spontaneous/evoked activity ratio had to be below 0.7 to avoid periods approaching emergence from anesthesia, and 2) and the population spike rate had to exceed 13 spikes s^{-1}, which excludes blocks suppressed under an overly deep plane and ensures sufficient spiking for a stable AR fit. Gray-screen control animals, which have no evoked response against which to define the ratio, were included under the rate criterion alone (see Fig. 3 analysis, below).

Spike-time shuffle control. To establish a white-noise upper bound on d_2 empirically, we computed a shuffled control (d_2^{shuf}) in which each neuron's spike count within a window was preserved while its spike times were redrawn independently and uniformly at random across the window. Shuffled single-neuron trains were recombined into a population time series and d_2 was recomputed as for intact data. This preserves each neuron's firing rate while destroying temporal correlations across the population, driving d_2^{shuf} toward the analytic white-noise value $1/\sqrt{10}$ (mice, Fig. 1C; ferrets, Fig. 3D).

Proximity to criticality (d_2): field potentials (humans)

Clinical stereo EEG resolves mesoscale LFP rather than single-unit spikes. We therefore fit the AR model to a continuous field-derived signal: the amplitude envelope of the high-gamma band, which serves as a proxy for local multi-unit bursting[75,76] and whose long-range temporal correlations are preserved above the recording hardware's high-pass cutoff. Within each preprocessed 180 s window (above), a single frontal contact was band-pass filtered to the gamma band (70–250 Hz; 4th-order Butterworth, upper edge capped just below the 250 Hz Nyquist), and the analytic amplitude envelope was obtained by Hilbert transform and smoothed with a 100 ms Gaussian kernel. The envelope was z-scored (rendering d_2 invariant to absolute signal amplitude) and fit with an order-10 AR model by Yule–Walker maximum-likelihood estimation. d_2 was computed by Eq. (2). Each patient's d_2 is the mean across that patient's 180 s windows; a 16-way aggregator sweep confirmed the mean is optimal for the effect and that the result is robust to this choice.

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Baseline d_2 setpoint and its stability (Fig. 1E, S3)

Because d_2 varies systematically by arousal state[40], baseline analyses focused on wake to estimate each animal's homeostatic setpoint; sleep-dominated windows (awake $< 90\%$) were excluded. For the trajectory display (Fig. 1F), each animal's wake d_2 samples were plotted against time since the start of baseline and smoothed with a 6 h Gaussian kernel (± 3 h); although the abscissa advances in clock time, only wake samples contributed and sleep periods contain no data points.

To test whether the setpoint is a durable individual trait, we partitioned the variance of the 5 min d_2 estimates. The single-window reliability was the one-way random-effects intraclass correlation $\mathrm{ICC}(1,1)$ (animal identity as the grouping factor); the reliability of the per-animal mean was $\mathrm{ICC}(1,k)$, corrected for the autocorrelation of successive windows by computing the effective number of independent windows (n/τ, $\tau \approx 5$ windows). Because consecutive windows are autocorrelated, the between-animal effect was tested by a one-way ANOVA on independent 3 h blocks with a label-permutation null, and reproducibility was corroborated by a split-half test–retest correlation (setpoint from the first vs. second non-overlapping half of baseline) with the Spearman–Brown full-length correction. Operational separability of animals was assessed by (i) the pairwise area under the ROC curve (AUC) for discriminating each pair of animals from single-window d_2, related to the difference in setpoints by Spearman correlation, and (ii) an 18-way cross-validated linear-discriminant classifier of animal identity from d_2, evaluated against a label-permutation null.

Prey-capture scoring and learning-rate quantification (Fig. 1H, S5)

Capture-time scoring. The time to capture a red runner on each trial was scored by an expert rater. A hunt began when the roach was dropped into the arena and ended when the mouse bit off the roach's head; when head-biting was not observed, the end was scored when the mouse settled into a stationary eating posture. Capture times were resolved to the nearest second and bounded above at 1,800 s (the 30 min timeout), with the exception of the pre-trial roach, which was available overnight; the pre-trial elapsed time is the true end $-$ start interval and capture-vs-failure was read from whether the cockroach was dead or consumed at lights on. Capture times were log-transformed for all analyses.

Learning-rate quantification. Prey-capture learning speed was estimated from trial-by-trial capture times. Because learning can be conceptualized either as continuous improvement toward an asymptote[112,113] (exponential model) or as skill acquisition followed by a plateau[114] (breakpoint model), we extracted complementary metrics using two fits. For the **exponential model**, log capture time across trials n was fit to $y = A\,e^{-kn} + C$ by nonlinear least squares, with the decay rate k constrained to $[0, 0.5]$; larger k indicates faster learning. To guard against timed-out trials driving the estimate, a second fit excluded timeout trials ($\geq$ 1,800 s), yielding a second rate k_2. For the **breakpoint model**, log capture time was fit to a piecewise-linear hinge with a flat tail, $y = \alpha - \beta \min(n - \tau, 0)$, in which performance improves with slope β up to breakpoint τ and is constant thereafter; τ was selected by exhaustive search over candidate trials ($\tau \geq 3$), minimizing the residual sum of squares. This yielded four duration-based metrics: breakpoint slope β, learned-trial number τ, exponential decay rate k, and exponential decay rate excluding failures k_2. Because τ has the opposite sign convention, metrics were sign-corrected so that higher always indicates faster learning, standardized, and collapsed by principal-component analysis (PCA). The first principal component (PC1) explained 89.2% of the variance and served as the prey-capture learning-speed measure (Fig. S5F). Metric concordance was confirmed by Kendall's W.

Early- and late-phase controls. Day-1 learning (Fig. 1G) used the median capture time over each animal's first hunting day. Naive ability (Fig. S5C) used the first hunting trial, and pre-

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learning behavior (Fig. S5B) used the overnight pre-trial roach (continuous latency among animals that caught it; a binary caught/failed outcome otherwise). Asymptotic performance was indexed by the median capture time over each animal's final ten trials, the fitted exponential asymptote C, and the mean late-phase latency; in-task locomotion (mean, peak, and total DLC velocity during hunts) was tested as an athleticism control.

Ladder-cross preprocessing and behavioral quantification (Fig. 2)

Touch preprocessing (Fig. 2B). Mice traversed the ladder in both directions; to make crosses directionally comparable, rungs were re-indexed so that rung 1 was always the first rung on the entry side and rung 10 the exit side. Rung contacts were detected by the MPR121 capacitive sensors, which register paw contacts but not tail contacts. To suppress within-contact sensor jitter, two touches on the same rung separated by less than 50 ms were merged into one, under the assumption that an animal cannot physically touch, lift, and re-touch a rung within 50 ms. Only crosses shorter than 60 s were retained. All quantitative ladder measures were derived from the MPR121 touch time series; video was used only to generate human labels for the cross-type classifier and contributed no measurement.

Trot detection by hidden Markov model (Fig. 2C-E). To quantify the emergence of trot-like coordination, we fit a per-animal categorical hidden Markov model (HMM) to the sequence of rung-touch events[115]. The touch series for a cross was binned into 10 ms samples and represented as a binary pattern over a 5-rung window on each side of the ladder ($2\times5 = 10$ bits), yielding 2^{10} possible observation symbols; only touches lasting $\geq$ 100 ms were included. Five hidden states captured the canonical trot cycle: the two diagonal stances (front-left/back-right and front-right/back-left), the two transient three-paw configurations bridging them, and an "other" state absorbing non-trot patterns (stalls, rearing, sniffing). The emission matrix was initialized by mapping each binary pattern to the state whose paw configuration it most closely matched; the transition matrix was initialized to favor the cyclic trot progression ($1\to2\to3\to4\to1$; transition 0.2, self-transition 0.7, 0.1 into "other"). Each per-animal model was then refined with `hmmlearn`[116] on that animal's last 300 crosses labeled "walk" or "trot" by the XGBoost classifier (below), matching each model to its own late-session gait statistics. After fitting, the hidden-state sequence for every cross was decoded, and a cross was classified as a trot if the product of posterior probabilities along the decoded trot-phase path exceeded $0.8^8 \approx 0.167$ (80% confidence at each of the $\sim$8 phase transitions of a typical cross).

Behavioral classification of ladder crosses. To stratify the diverse behaviors mice display on the ladder, we trained an XGBoost classifier[105] to assign each cross to one of six mutually exclusive categories: u-turn, stall-and-sniff, vertical exploration (rearing), walk, trot, and bound; the walk/trot/bound gait definitions follow Bellardita and Kiehn[117]. The classifier was trained on 2,328 manually labeled crosses from 8 animals. For each cross, 33 scalar and binary features were extracted, including cross duration, number of rungs touched, touch-time covariance, and left–right limb symmetry. Labeled data were split 70/30 into training and held-out test sets; random over-sampling corrected class imbalance in the training set, and a multi-class classifier was fit (max depth 5, 200 estimators), achieving 85% balanced accuracy on held-out data. The manifold analyses (Fig. 2I-J; Fig. S8C-D) used these XGBoost gait labels.

Manifold embedding of ladder crosses (Fig. 2I-J) To analyze ladder crosses at the population scale, we embedded them in a low-dimensional manifold. Because crosses vary in their number of touches and a rung may be touched multiple times (especially early in learning), fixed-length-vector methods (PCA, UMAP) are poorly suited. We instead treated each cross as a sequence

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of touch events, computed pairwise distances by sequence alignment, and embedded the distance matrix in two dimensions by multidimensional scaling (MDS)[55,118]. Each cross was converted to an ordered sequence of touches, each touch described by four attributes (start time relative to cross onset, duration, body side, rung position 1–10); only touches $\geq$ 100 ms and crosses with more than four such touches were used (touches per cross: mean $\pm$ s.d. = 10.61 ± 5.14; range 5–150). Pairwise distances used Needleman–Wunsch (NW) global alignment[119]: the substitution cost between two touches was the sum of a 2-point penalty for opposite sides, the absolute rung-position difference, twice the relative duration difference, and four times the relative start-time difference; insertions/deletions incurred a gap cost of 16. MDS was run on the resulting per-animal distance matrix with 10 random initializations. For Fig. 2J and S8C-D, each animal was restricted to its first 200 crosses, and we computed the convex-hull area of the full embedding (the *all*-manifold area) and of the walk/trot subset (the *walk/trot*-manifold area); their ratio is the exploration/consolidation index (Fig. 2J).

Because metric MDS reproduces the absolute values of the input distances rather than their ranks, and all pairwise distances were computed with the same NW cost function and parameters, the resulting convex-hull areas share a common scale ($NW distance^2$) across animals. This makes cross-animal comparison of manifold size valid without post-hoc normalization. Scale-normalizing methods such as UMAP would not support this comparison, as they standardize local neighborhood structure and produce embeddings on an arbitrary per-animal scale.

Ferret plasticity and d_2 (Fig. 3)

For each ferret and exposure block, baseline d_2 was computed from the assessment epoch immediately preceding exposure (above), and plasticity was the change in cell-averaged $1 - \mathrm{DCV}$ across that block. This yielded up to three (d_2, plasticity) points per animal. Points entered the correlation if the block passed the population-rate and spontaneous/evoked gates (non-control animals) or the rate gate alone (control animals), giving $n = 33$ blocks from 14 kits. Because the control-condition animals followed the same $d_2 \to$ plasticity relation, including them tightened the fit; controls-included is the reported set. Because plasticity is an unbounded, signed change (unlike the floored, saturating mouse learning outcomes), the primary statistic here is Pearson r (with Spearman ρ and a permutation null as robustness checks). Pseudoreplication from multiple blocks per animal was addressed by a linear mixed-effects model, plasticity $\sim d_2 + (1 \,|\, \mathrm{animal})$, and by collapsing to one point per animal ($N = 14$); the spike-time shuffle control (above) served as a negative control. The training-time increase in selectivity (Fig. 3B-C) was quantified by a linear mixed-effects model, direction selectivity $\sim$ exposure time $+ (1 \,|\, \mathrm{animal})$, over the motion animals.

Minimal recurrent model relating d_2 to learning rate (Fig. 3F-H)

Temporal credit assignment in a recurrent network trained by backpropagation through time is governed by a product of recurrent Jacobians that, for dynamics dominated by a single stable real mode $\lambda_* \equiv \mu$, scales as μ^k over a delay of $k = t - s$ steps[67,68]. To obtain a closed-form relationship between distance to criticality and learning rate, we reduced the network to this dominant mode and analyzed learning of a delayed input–output association exactly.

Dominant-mode reduction and d_2. Retaining only the slowest mode, the hidden dynamics reduce to the scalar recursion

$$h_t = \mu\, h_{t-1} + I_t, \qquad 0 < \mu < 1, \tag{3}$$

a first-order autoregressive (AR(1)) process. For an AR(p) model, $d_2 = |1 - \sum_{\tau=1}^{p} \phi_\tau| / \sqrt{p}$ (Eq. (2)); for AR(1) ($p = 1$, $\phi_1 = \mu$) this is simply

$$d_2 = 1 - \mu, \tag{4}$$

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so proximity to criticality ($\mu \to 1$) corresponds exactly to $d_2 \to 0$. The credit-assignment gain over a delay of k steps is therefore

$$G(k) = \mu^k = (1 - d_2)^k, \tag{5}$$

which decays geometrically with delay at a rate fixed entirely by d_2 (Fig. 3F).

Exact learning dynamics. Each trial delivers a fixed input I at trial onset; the hidden state then decays freely, $h_\tau = \mu^\tau I$, and is read out k steps later, $h_k = \mu^k I$, where k is the credit-assignment horizon of the task. A linear readout produces the output $o_m = w_m\, h_k = w_m\, \mu^k I$ on trial m, with squared-error loss $\ell_m = \frac{1}{2}(y^* - w_m \mu^k I)^2$ against a fixed target y^*. Gradient descent on w with step size α gives

$$w_{m+1} = w_m + \alpha\big(y^* - w_m \mu^k I\big)\mu^k I, \tag{6}$$

with unique optimum $w^* = y^*/(\mu^k I)$ and loss floor $\ell^* = 0$ for every network, independent of d_2. Writing the error as $e_m = w_m - w^*$, the update contracts linearly,

$$e_{m+1} = \big(1 - \alpha \mu^{2k} I^2\big)\, e_m \implies \ell_m = \big(1 - \alpha I^2 (1 - d_2)^{2k}\big)^{2m} \ell_0, \tag{7}$$

where ℓ_0 is the initial loss and the exponent $2m$ follows from the loss being quadratic in e_m. Equation (7) gives the family of learning curves in Fig. 3G.

Trials to criterion. Defining learning as reaching a fixed loss tolerance $\ell_M \le \epsilon$ and solving Eq. (7) for M gives the number of trials required,

$$M(d_2, k) = \frac{\ln(\epsilon/\ell_0)}{2\,\ln\big[\,1 - \alpha I^2 (1 - d_2)^{2k}\,\big]} \approx \frac{\ln(\ell_0/\epsilon)}{2\,\alpha I^2\,(1 - d_2)^{2k}} \propto (1 - d_2)^{-2k}, \tag{8}$$

the small-step approximation ($\ln(1 - z) \approx -z$) holding whenever $0 < z \equiv \alpha I^2 (1 - d_2)^{2k} \ll 1$. Equation (8) is the surface in Fig. 3H; it decreases monotonically as $d_2 \to 0$ and steepens with the credit horizon k, so the penalty for being off-critical compounds with task difficulty.

Figure generation. All panels used $\alpha = 0.05$, $I = 1$, $\ell_0 = 1$, and $\epsilon = 0.05$. Fig. 3F plots the credit gain (Eq. (5)) against delay $k \in [0, 60]$; Fig. 3G plots the learning curves (Eq. (7)) against trial $m \in [0, 500]$ at fixed $k = 10$; both show eight values of d_2 evenly spaced over $[0.05, 0.25]$. Fig. 3H plots $\log_{10} M$ (Eq. (8)) over $d_2 \in [0.001, 0.30]$ and $k \in [10, 100]$. To avoid catastrophic cancellation when z is small, $\ln(1 - z)$ was evaluated as `log1p`$(-z)$.

Human d_2 versus IQ (Fig. 4)

The analysis was between-patient: one d_2 per patient versus one full-scale IQ, paralleling the mouse learning-rate design. A window-level mixed model with a patient random intercept is degenerate here because IQ is constant within patient (the random intercept absorbs the entire between-patient effect), so it was not used. The primary model was a partial correlation of d_2 with IQ adjusting for demographic covariates (sex and age), which are uncorrelated with d_2 and therefore act as suppressors; the zero-order Pearson correlation is also reported. Significance of the adjusted effect was assessed by a Freedman–Lane residual-permutation test. Model form was checked by BIC over monotonic candidates (linear, exponential, power, reciprocal), which selected linear. Robustness was established by a 16-way aggregator sweep (window-to-patient collapse), leave-one-patient-out, and a clinical-sensitivity model additionally adjusting for autism, seizure burden, and resection status (which are entangled with d_2 and attenuate the estimate; reported as a supplement, not the primary).

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Baseline control-feature (specificity) panels (Fig. 1J; Figs. 2E,H; Fig. 3E; Fig. 4E)

To test whether d_2's predictive power is specific, in each figure d_2 was competed against a panel of additional baseline features, each entered as a standalone univariate predictor of the same outcome. For mice, the panel comprised 20 control features spanning spike-rate/regularity statistics (mean RSU firing rate; CV, CV2, LV, LVR; Fano factor at 100 and 200 ms; Lempel–Ziv complexity of the 15 ms-binned population train), inter-spike-interval statistics (min, max, median, mode, and s.d. of the ISI), normalized LFP band power (delta, theta, alpha) and band ratios (gamma/alpha, gamma/beta), and animal-level covariates (DLC movement, wake proportion); with d_2 this is 21 neural/behavioral features. For ferrets, only the 13 spike-train–computable features were available (behavioral and LFP features have no anesthetized equivalent). For humans, d_2 was competed against 13 *amplitude-invariant* spectral features only—relative band powers, band ratios (θ/β, γ/θ, high-γ/θ), spectral entropy, and the envelope coefficient of variation—because absolute sEEG amplitude is set by non-physiological factors (electrode geometry, impedance, reference, gain) and is not comparable across patients; absolute-amplitude features (signal std/var/RMS, absolute and log band powers, envelope mean/std/var) form a single collinear nuisance factor and were excluded.

Each panel bar is the univariate effect size against the figure's outcome, using the statistic native to that figure: Spearman ρ^2 for the mouse learning outcomes (a bounded, saturating quantity), Pearson r^2 for ferret plasticity (an unbounded signed change), and the sex+age-adjusted partial r^2 for human IQ. Significance was read against a permutation null built by shuffling the outcome relative to each feature. Because the ρ^2 null under the null hypothesis depends only on n (continuous metrics, no ties), a single large shared permutation null (200,000 draws) was built per n and used for both every feature's permutation p and the two reference lines drawn on each panel: the null median (chance) and the 95th percentile (the per-test $p < 0.05$ boundary).

No multiple-comparisons correction. No multiple-comparisons correction was applied to the specificity panels. These panels are confirmatory of a *negative* prediction—that the effect should be specific to d_2—not discovery tests. A spurious hit in a control feature can only *weaken* the specificity claim, never strengthen it, so FDR/Bonferroni would perversely penalize thoroughness (adding arbitrary controls would lower q for the primary without changing its predictive power). Significance is therefore read against the chance and 95th-percentile reference lines and the permutation null.

Relationship of d_2 to elementary signal statistics (Fig. 1D; Fig. S2)

To test whether d_2 merely re-expresses a simpler feature such as variance or slow-frequency power, we characterized the full space of stationary second-order autoregressive [AR(2)] processes—the lowest-order model that supports the scale-invariant dynamics used to define criticality[40]. AR(2) processes are specified by $(\phi_1, \phi_2, \sigma^2)$, with the stationary region forming a bounded triangle in the (ϕ_1, ϕ_2) plane and the critical family lying along $\phi_1 + \phi_2 = 1$; there, $d_2 = |1 - (\phi_1 + \phi_2)|/\sqrt{2}$. The quantities of interest have closed forms over this plane: the stationary variance is

$$\mathrm{Var}(x) = \frac{\sigma^2\,(1-\phi_2)}{(1+\phi_2)\left[(1-\phi_2)^2 - \phi_1^2\right]}, \tag{9}$$

and the power spectral density is $S(f) = \sigma^2 / \left|1 - \phi_1 e^{-i\omega} - \phi_2 e^{-2i\omega}\right|^2$ with $\omega = 2\pi f \Delta t$. To match the d_2 analysis, spectral quantities used the same bin width, $\Delta t = 15$ ms, with relative delta power defined as the fraction of total power in 0.5–4 Hz. Evaluated over the entire stable region, d_2 increases monotonically with distance from the critical line, whereas variance and slow-frequency spectral content do not, and each conventional descriptor's relationship to d_2 is local—its sign and magnitude shift, and can invert, across the plane. Thus no single statistic tracks the critical boundary coherently across animals the way d_2 does.

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Single-neuron and population control metrics

The control-panel features were computed from the same RSU populations and 5 min wake windows used for d_2. The Fano factor was computed per neuron and averaged across neurons to give one value per animal per window. Inter-spike-interval regularity metrics (CV, CV2, LV, LVR) were likewise restricted to RSUs, computed per neuron, and averaged per window[23]. Population Lempel–Ziv (LZ) complexity followed the binarization scheme of Abásolo et al.[120]: population spike trains were converted to binary sequences at 500 Hz (2 ms bins, 1 if ≥ 1 spike, else 0) at the 15 ms binning matched to d_2, and LZ complexity was estimated from these binary population sequences. Normalized LFP band powers and band ratios were computed from the baseline LFP over the same windows.

Statistics

Analyses were performed in Python (`numpy`, `scipy`, `statsmodels`, `scikit-learn`, `hmmlearn`, `xgboost`). Linear mixed-effects models (ferret plasticity and selectivity-vs-time) were fit with `statsmodels` (`mixedlm`, restricted maximum likelihood), with fixed-effect significance assessed by the Wald z test.

Primary statistics by figure. For the mouse learning outcomes, the primary statistic is Spearman's ρ (rank), because learning outcomes are bounded and saturating (capture time cannot fall below zero and decays to a floor); all mouse specificity panels square Spearman ρ^2. For ferret plasticity, an unbounded signed change, the primary statistic is Pearson's r, with Spearman ρ as a robustness check; a linear mixed-effects model with animal as a random effect provides the inferential p-value against block pseudoreplication. For the human analysis, the primary statistic is the demographic-adjusted (sex, age) partial correlation. Pearson/linear fits and BIC model-selection fits are reported alongside rank statistics where relevant but are not the headline result.

Permutation nulls. For key correlations we report a permutation p-value obtained by shuffling one variable relative to the other, recomputing the correlation under each shuffle, and taking the proportion of shuffled correlations whose absolute value met or exceeded the observed value. Nulls used 5,000 shuffles for individual correlations and 200,000 for the shared specificity-panel nulls. The human adjusted effect used a Freedman–Lane residual permutation.

Model selection. Where a functional form was fit to a scatter (e.g., the ladder exploration/consolidation ratio, Fig. 2; the human d_2–IQ relationship, Fig. 4), the model was selected by the Bayesian Information Criterion (BIC) over a set of *monotonic* candidates (linear, exponential, reciprocal, power); quadratic and other non-monotonic models were excluded, as a U-shape has no justification under the criticality hypothesis and overfits at these sample sizes. For the ladder ratio, an unbounded and magnitude-meaningful quantity, BIC was scored in log space (minimizing squared error in $\log_{10}$) to prevent a single high-leverage animal from dominating the raw-space fit, and Pearson r on log–log axes is reported; BIC selected a power law.

Multiple comparisons. No multiple-comparisons correction was applied to the specificity panels, for the reasons given above (they are confirmatory of a negative prediction). Reported p-values are two-tailed and, unless noted, uncorrected.

Data availability

Source data and derived tables supporting the findings of this study are available from the corresponding authors upon reasonable request. Human sEEG data are coded clinical data and are subject to the governing IRB and data-use agreements. Ferret data (Ritter et al.[60]) are available from S.D.V.H.

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Code availability

Analysis code is available at `https://github.com/hengenlab`.

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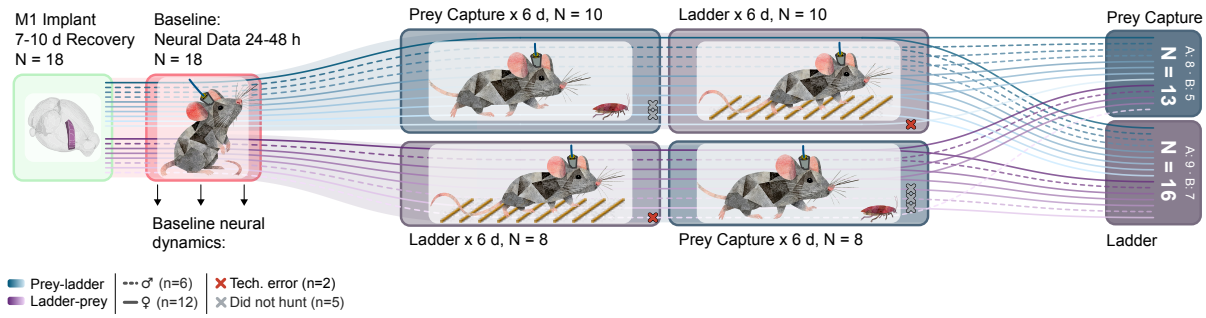


Figure S1: Experimental pipeline and per-animal task assignment. Eighteen mice (6 male, 12 female) received chronic right M1 microelectrode implants, recovered for 7-10 days, and underwent 24–48 h of task-free baseline neural recording before completing two counterbalanced 6-day behavioral tasks: prey capture (roach hunting, ∼3 h/morning) and ladder crossing (continuously available in the homecage). Ten animals performed prey capture first (Group A, blue) and eight performed ladder crossing first (Group B, purple); solid lines trace individual females and dashed lines trace individual males across the pipeline. Downstream analysis datasets (right) comprise 16 animals with valid ladder-crossing data and 13 confirmed roach hunters; 11 animals contribute to both. Exclusions are marked with ×: two animals lost ladder data to equipment failure but retained valid hunting data (red, "technical ladder failure"), and five animals never hunted a roach and were excluded from the prey-capture analysis only (gray, behavioral non-consolidation unrelated to d_2).

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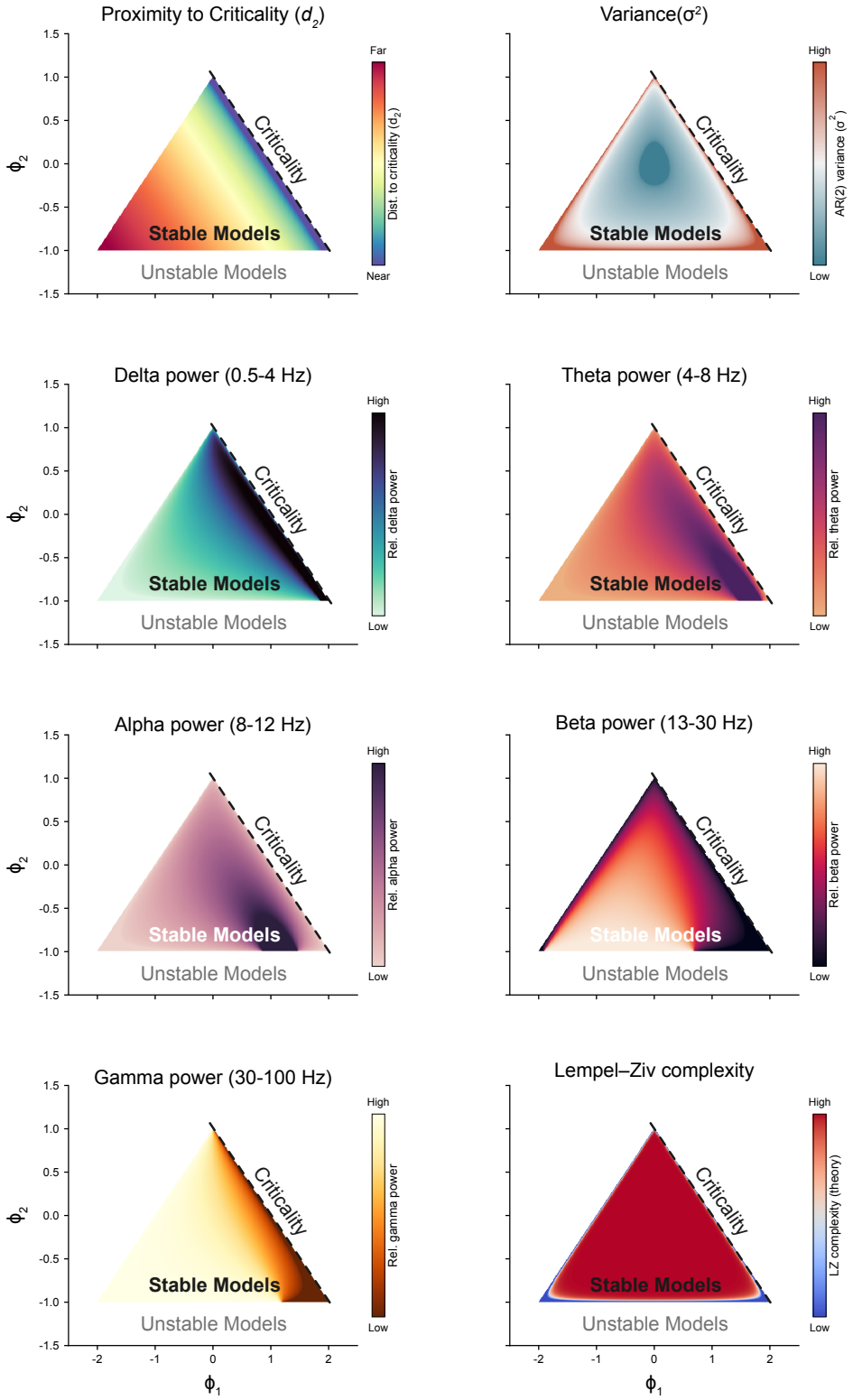


Figure S2: Features of neural activity exhibit distinct patterns when considered across a space of possible dynamics. Each panel maps a candidate metric across the stable region of the AR(2) parameter space (ϕ_1–ϕ_2 plane; critical boundary $\phi_1 + \phi_2 = 1$, dashed), colored by that metric's value. Unlike d_2 (top left), which increases monotonically with distance from the critical line everywhere in the space, other comparator descriptors—variance (σ^2), relative band power (δ, θ, α, β, γ), and Lempel–Ziv complexity—tracks d_2 only locally, with a relationship whose sign and magnitude shift across the plane.

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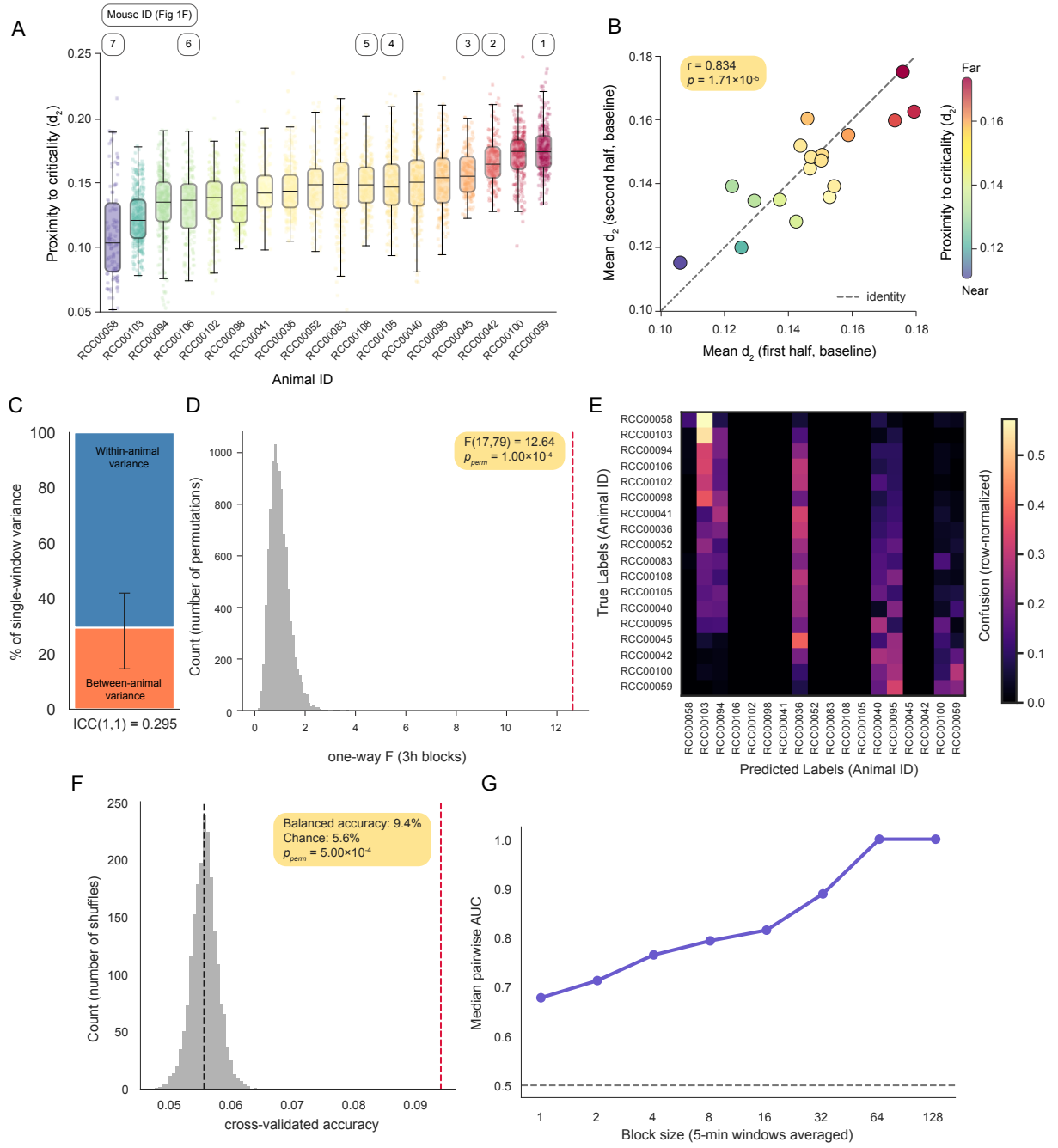


Figure S3: d_2 is stable and discriminable as an individual setpoint. **(A)** Per-animal distribution of baseline d_2 (300-s windows, M1, wake), animals ordered by mean d_2; boxes show median and IQR, points are individual windows, color encodes mean d_2 ($N = 18$). **(B)** Split-half test–retest: mean d_2 from first vs. second half of each animal's baseline (Pearson $r = 0.834$, $r^2 = 0.695$, $p = 1.71 \times 10^{-5}$). **(C)** Variance decomposition of single-window d_2: between-animal variance is only 29.5% (ICC(1,1)= 0.295, 95% CI [0.147, 0.420]). **(D)** One-way ANOVA for animal identity (F(17,79)= 12.64) falls far outside a 3-h-block label-shuffle permutation null (0/10,000 permutations $\geq$ observed, $p_{\text{perm}} < 10^{-4}$), confirming the animal effect is not noise. **(E)** Confusion matrix from 18-way linear discriminant classification of animal identity from single-window d_2 (2-fold interleaved cross-validation within animal; row-normalized). **(F)** Balanced classification accuracy against a label-shuffle null (2,000 shuffles). **(G)** Pairwise discriminability (cross-validated AUC) rises as 5-min d_2 windows are averaged into longer blocks, from a median AUC of 0.68 at one window to 1.00 by $\sim$5.3 h of integration (64 windows).

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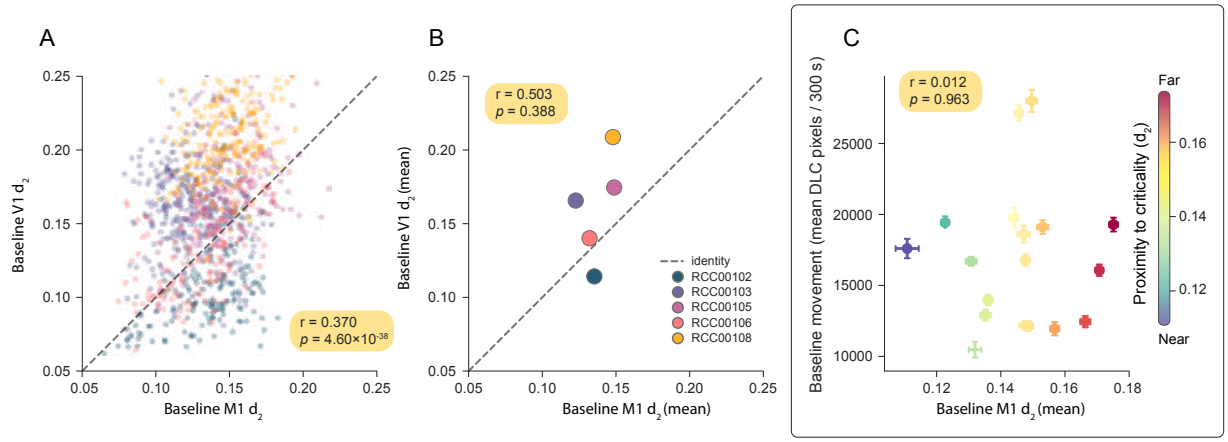


Figure S4: Baseline proximity to criticality is not a global, physiological set-point, nor is it a trivial proxy for baseline motor vigor. (A) For $N = 5$ mice where concurrent baseline data were recorded from M1 and V1, there is a weak correlation between M1 and V1 d_2 when all non-overlapping epochs are considered (Pearson $r = 0.37$, $p = 4.6 \times 10^{-38}$, $n = 1131$ blocks of wake). **(B)** When the mean d_2 set-points for M1 and V1 from these same mice are considered, there is no correlation (Pearson $r = 0.50$, $p = 0.39$, $N = 5$). **(C)** When baseline M1 d_2 is considered as a predictor for the amount of running an animal does in the homecage, there is no correlation (Pearson $r = 0.01$, $p = 0.96$; Spearman $\rho = -0.05$, $p = 0.84$; $N = 18$ animals).

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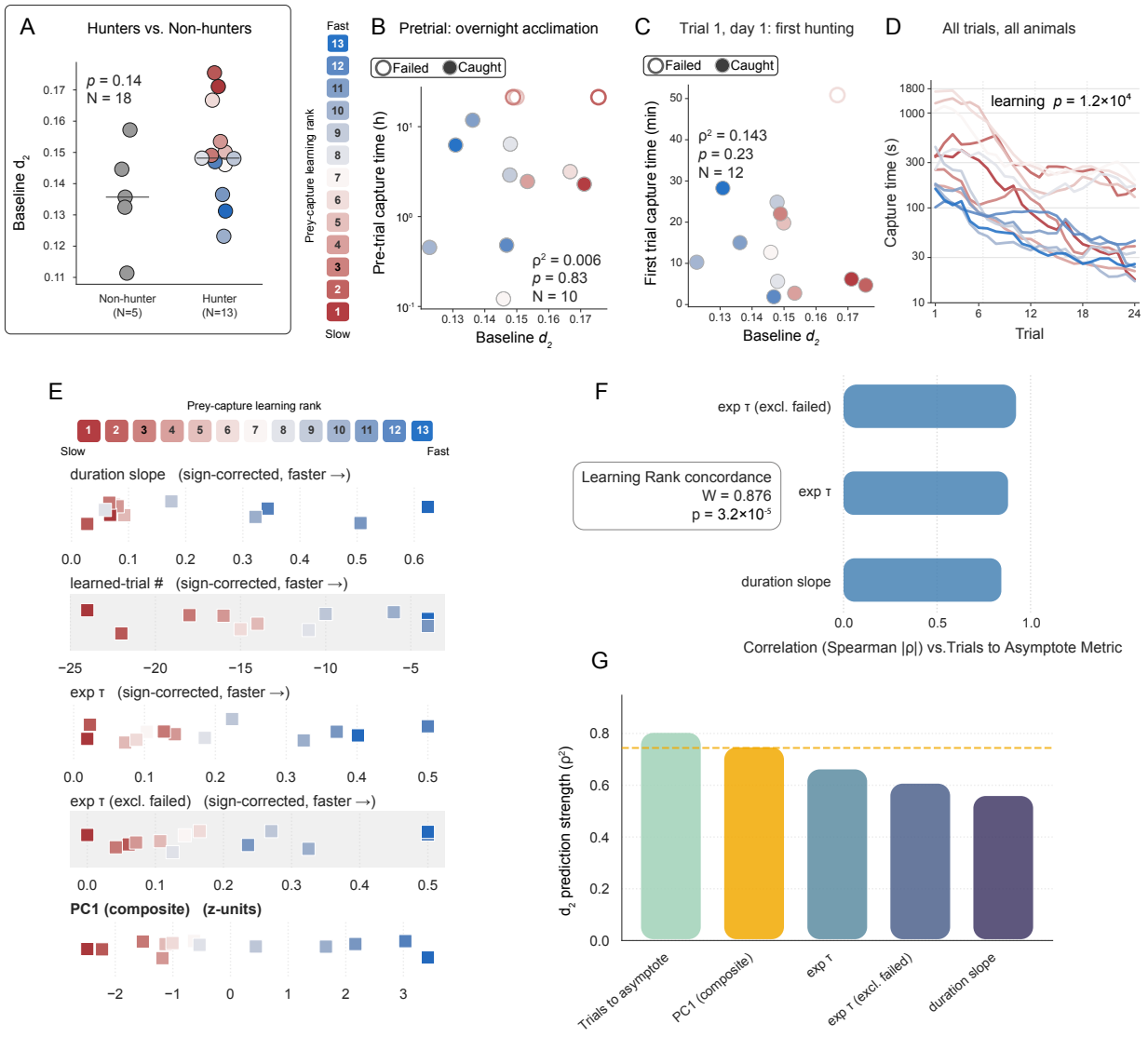


Figure S5: Animals show learning in a prey capture paradigm, and initial performance is not predicted by proximity to criticality. **(A)** Baseline d_2 does not differ between animals that hunted (Hunter, $N = 13$) and animals that never made a successful strike (Non-hunter, $N = 5$; Mann–Whitney $U = 48.0$, $p = 0.14$). Non-hunters are excluded from B–G as behaviorally non-interpretable (motivational or sensory, not slow learning), not because of their d_2. **(B)** Capture time on the overnight acclimation roach, before any learning trial, is unrelated to baseline d_2 ($n = 10$ that caught it; Spearman $r^2 = 0.11$, $p = 0.83$); open circles never captured this roach. **(C)** Capture time on the first real trial (day 1, trial 1) is likewise unrelated to baseline d_2 ($n = 12$ caught; $\rho^2 = 0.143$, $p = 0.23$), so d_2 does not track innate hunting ability at first exposure. **(D)** Every animal improves over the 24-trial course regardless of d_2 (6-trial rolling-mean capture time; 13/13 animals show a negative within-animal Spearman correlation between capture time and trial number, sign test $p = 1.2 \times 10^{-4}$) – the d_2 relationship emerges only once trials accumulate. **(E)** Distributions of the four constituent duration-learning metrics (duration slope, trials-to-asymptote, exponential time constant τ, and τ excluding failed hunts, each sign-corrected so higher = faster learner) and their first principal component (PC1, the composite learning score used throughout), animals colored by PC1 rank – the four metrics order animals concordantly. **(F)** Pairwise rank concordance (Spearman $|\rho|$) of each metric against the reference (trials-to-asymptote) is high (Kendall's $W = 0.876$, $p = 3.2 \times 10^{-5}$), so the composite is not built from disagreeing metrics. **(G)** Baseline d_2's predictive power (Spearman ρ^2) for each metric and for PC1 (gold); PC1 ranks 2nd of five for d_2 prediction ($\rho^2 = 0.744$), mid-range among its own constituents rather than an artificially inflated combination.

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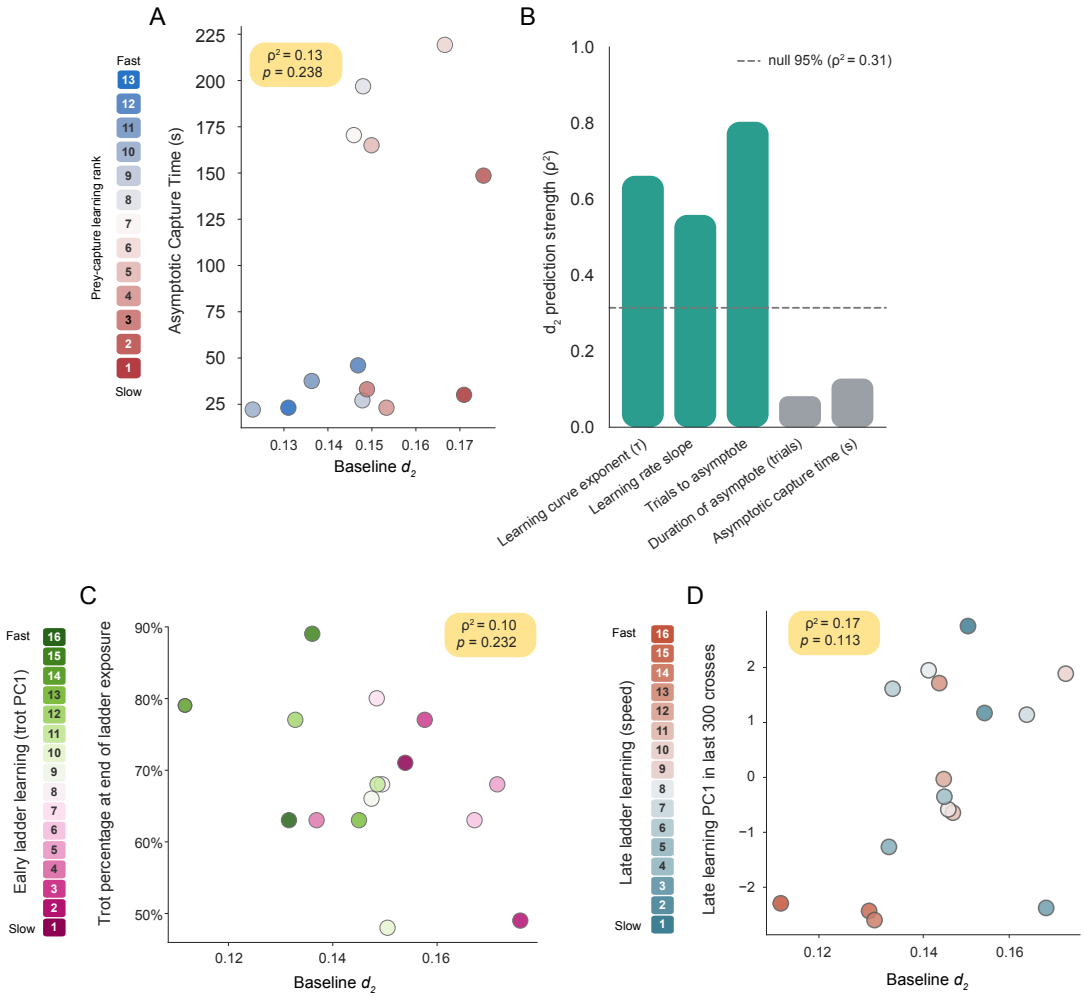
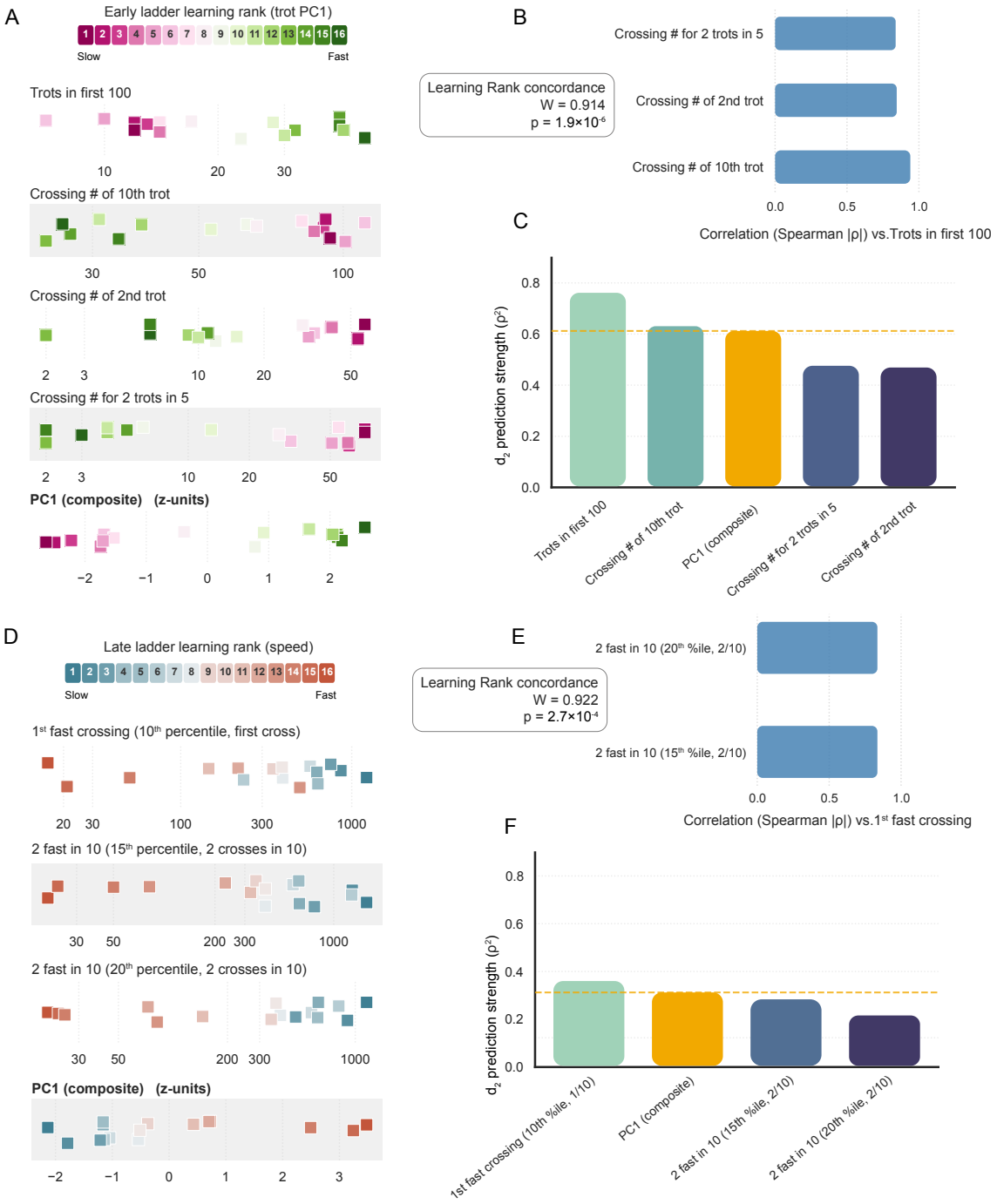


Figure S6: Proximity to criticality does not predict behavioral asymptote in prey capture or ladder learning. (A) There was no correlation between baseline d_2 and median capture time across each animal's final 10 trials (its behavioral asymptote/ceiling) (Spearman $\rho = 0.36$, $\rho^2 = 0.13$, $p = 0.234$; $n = 13$ hunters). **(B)** Compared to its prediction strength for behavioral-asymptote measures, baseline d_2 was far more predictive of learning-*rate* measures: three rate metrics (exponential decay constant k: $\rho^2 = 0.66$, $p = 0.001$; duration slope: $\rho^2 = 0.56$, $p = 0.005$; learned-trial number: $\rho^2 = 0.80$, $p = 0.0002$) all exceeded two asymptote metrics (asymptote C: $\rho^2 = 0.08$, $p = 0.341$; median last-10 capture time: $\rho^2 = 0.13$, $p = 0.234$), $n = 13$ for all. The dashed *null* line marks the 95th percentile of a permutation null for Spearman ρ^2 (animal identity shuffled 5,000 times, matched to $n = 13$) — i.e., the effect size a metric would need to exceed by chance alone to reach an uncorrected $p < 0.05$ ($\rho^2 = 0.31$); asterisks mark the metrics clearing this threshold. **(C)** Baseline d_2 did not predict the endstage (ceiling) proportion of trots among the first 100 crossings of each animal's final 300 ladder crossings (Spearman $\rho = -0.32$, $\rho^2 = 0.10$, $p = 0.232$; Pearson $r = -0.46$, $r^2 = 0.21$, $p = 0.074$; $n = 16$). **(D)** Baseline d_2 likewise did not predict an endstage speed-milestone composite (PC1 of percentile on-ladder-duration milestones over the final 300 crossings; 99.6% variance explained) (Spearman $\rho = +0.41$, $\rho^2 = 0.17$, $p = 0.113$; Pearson $r = +0.45$, $r^2 = 0.20$, $p = 0.083$; $n = 16$).

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Figure S7: Animals learn and refine a repetitive motor sequence as they learn to cross a ladder. **(A–C)** Animals ($N = 16$) vary in how fast they learn to trot on the ladder. **(A)** Distributions of the four constituent trot-learning metrics (trots in the first 100 interactions, crosses to 10th trot, crosses to 2nd trot, and crosses until 2 trots occur within a 5-cross sliding window) and their first principal component (PC1, the composite learning score used throughout); animals colored by PC1 rank – the metrics order animals concordantly. **(B)** Pairwise rank concordance (Spearman $|\rho|$) of each metric against the reference (trots in the first 100 interactions) is high (Kendall's $W = 0.914$, $p = 1.9 \times 10^{-6}$), so the composite is not built from disagreeing metrics. **(C)** Baseline d_2's predictive power (Spearman ρ^2) for each metric and for PC1 (gold); PC1 ranks 3rd of five ($\rho^2 = 0.61$), mid-range among its own constituents rather than an artificially inflated combination. **(D–F)** Animals vary in how fast they approach an individual cross duration asymptote. **(D)** Distributions of the three constituent cross-duration-learning metrics (1st cross reaching that mouse's 10th percentile for cross duration, crosses until 2 of 10 reach the 15th percentile for speed, and crosses until 2 of 10 reach the 20th percentile) and their first principal component (PC1); animals colored by PC1 rank – the metrics order animals concordantly. **(E)** Pairwise rank concordance (Spearman $|\rho|$) of each metric against the reference (crosses until the mouse reaches its 10th percentile for speed) is high (Kendall's $W = 0.922$, $p = 2.7 \times 10^{-4}$), so the composite is not built from disagreeing metrics. **(F)** Baseline d_2's predictive power (Spearman ρ^2) for each metric and for PC1 (gold); PC1 ranks 2nd of four ($\rho^2 = 0.31$), mid-range among its own constituents rather than an artificially inflated combination.

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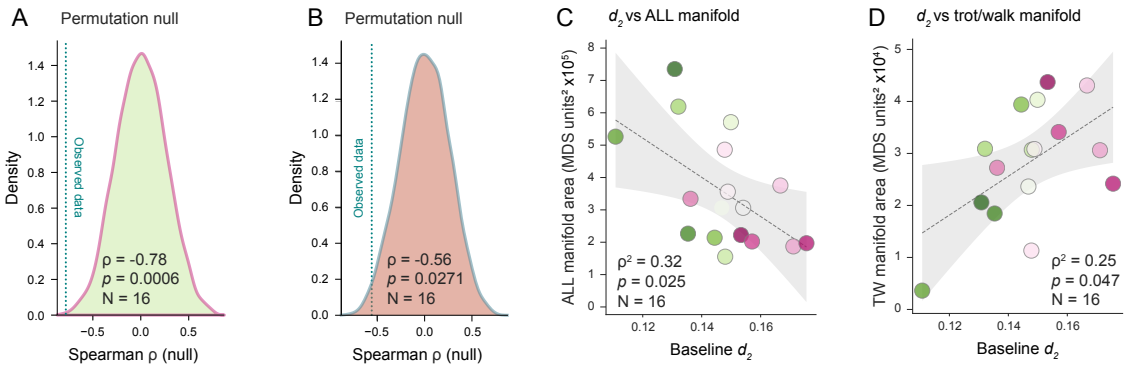


Figure S8: Mice learning the ladder show a robust prediction of early learning, late learning, and the geometric structure of behavior during learning. **(A)** Permutation null establishes that baseline d_2 predicts early ladder learning (early-learning composite, PC1) far beyond chance: the observed Spearman correlation ($\rho = -0.78$, teal dashed line) falls outside the extreme tail of a null distribution built from 200,000 label-shuffled draws ($p_{\text{perm}} = 0.0006$, $N = 16$ animals). **(B)** The same permutation test applied to late ladder learning (speed-milestone composite, PC1) shows baseline d_2 remains a significant predictor, albeit with a smaller effect size ($\rho = -0.56$, $p_{\text{perm}} = 0.027$, $N = 16$). **(C)** Baseline d_2 also predicts the geometric structure of behavior during learning: animals closer to criticality (lower d_2) traverse a larger overall movement repertoire, quantified as the area of the MDS-embedded behavioral manifold spanning all locomotor sub-types over the first 200 ladder crossings ($\rho^2 = 0.32$, $p = 0.025$, $N = 16$). **(D)** Conversely, the manifold restricted to purposeful trot/walk locomotion is smaller in more-critical animals and grows as d_2 increases ($\rho^2 = 0.25$, $p = 0.047$, $N = 16$) — so animals near criticality explore a broad overall repertoire built around a comparatively tight trot/walk core. In **C**–**D**, dots are colored by each animal's early-learning rank.

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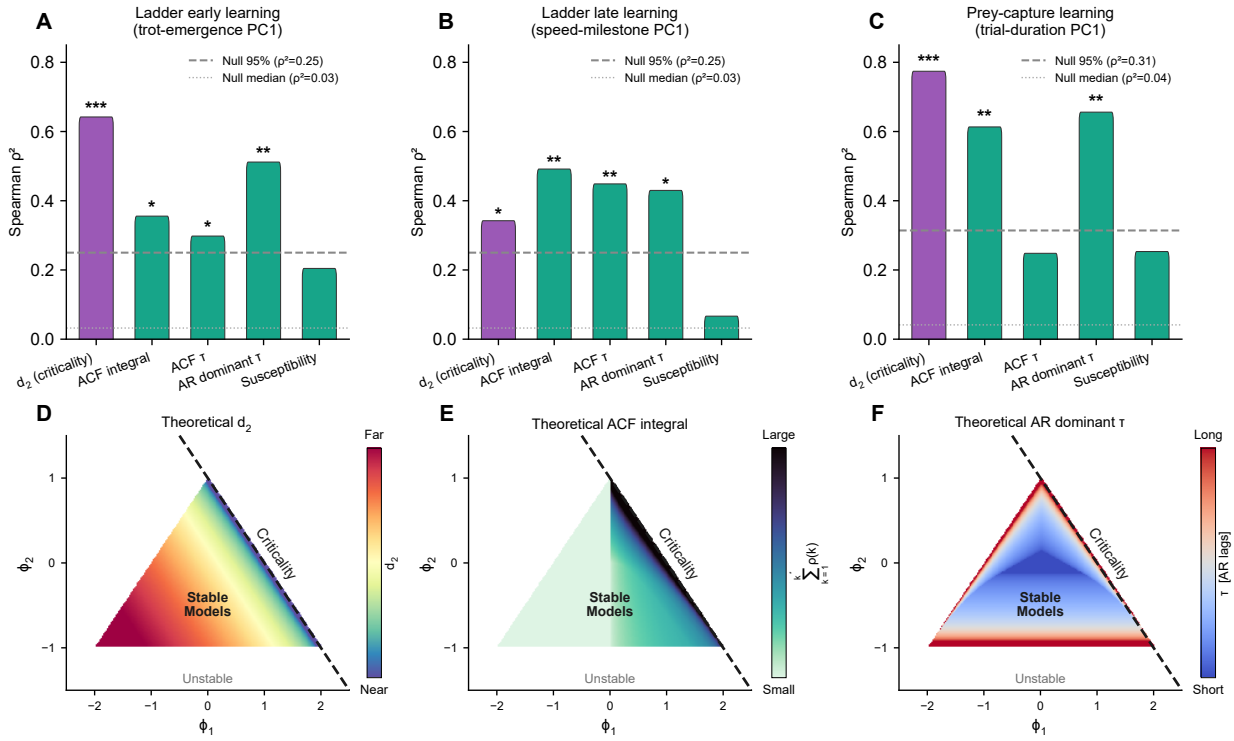


Figure S9: Autocorrelation timescales predict future learning and share dynamical origins with d_2 in AR(2) space. All metrics were derived from the 300 s wake baseline RSUs activity, using 15 ms population spike-count bins. d_2 (purple) is identical to that used throughout the paper. **ACF integral** is the area under the population activity autocorrelation function (ACF) up to its first zero-crossing. The ACF was fit with $Ae^{-t/\tau} + B$, and the asymptotic floor B was subtracted. **ACF** τ is the exponential time constant from the same fit. **AR dominant** τ is derived from the same AR(10) model used to compute d_2. AR dominant $\tau = -\Delta t / \ln|\lambda_{\max}|$, where $\lambda_{\max}$ is the dominant eigenvalue of the AR(10) companion matrix (from Yule–Walker); this is the memory timescale implied by the fitted model. **Susceptibility** is $N \cdot \mathrm{Var}(\rho_t)$, where ρ_t is the fraction of recorded cells that fired at least once (binarized) in bin t, following Eq. 2 of[121]. **(A–C)** Both d_2 and the autocorrelation measures (ACF integral, ACF τ, and AR dominant τ) predict ladder- and prey-capture learning, whereas susceptibility does not reach significance. **(D–F)** Theoretical behavior of each metric across the full space of stable AR(2) models (ϕ_1–ϕ_2 plane; dashed line, critical boundary $\phi_1 + \phi_2 = 1$). **(D)** The d_2 triangle is identical to that in Fig. 1D. The susceptibility triangle is not shown, as it matches the "Variance σ^2" heatmap in Fig. S2D. **(E)** The ACF integral (sum of $\rho(k)$ to the first zero-crossing) diverges at the critical line and falls to zero far from it. **(F)** The dominant autocorrelation timescale τ shows the same divergence at criticality.

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Supplementary Videos Path: `2026_chopra_zhong/database/supplemental_videos`

- Supplementary Video 1 (Early Crossing): `ladder_early_crossing.mp4`
- Supplementary Video 2 (trot, angle 1): `ladder_trot_angel1.mp4`
- Supplementary Video 3 (trot, angle 2): `ladder_trot_angel2.mp4`
- Supplementary Video 4 (trot at 0.25x speed, angle 2): `ladder_trot_0p25x_angle2.mp4`
- Supplementary Video 5 (u-turn): `ladder_uturn.mp4`
- Supplementary Video 6 (vertical exploration): `ladder_vertical_exploration.mp4`